SOME ASYMPTOTIC EXPANSIONS FOR PROLATE SPHEROIDAL WAVE FUNCTIONS

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I. Introduction and Summary of Results. 1.1 *Introduction.* For special values of the parameter χ, the differential equation

$$\frac{d}{dx}(1 - x^2)\frac{d\psi}{dx} + \left(\chi - c^2x^2 - \frac{m^2}{1 - x^2}\right)\psi = 0 \tag{1.1}$$

has a real continuous solution $\psi(x)$ that is bounded for all x. Except for a constant factor, this solution is unique. Here, as throughout this paper, c is an arbitrary positive number and m is a nonnegative integer. For simplicity of notation, we shall generally suppress the explicit dependence of other quantities on c.

Denote the special values of χ for which these bounded continuous solutions exist by $\chi_{m,n}$, $n = m, m + 1, \cdots$. These eigenvalues can be labelled so that $0 < \chi_{m,m} < \chi_{m,m+1} < \chi_{m,m+2} < \cdots$. The corresponding solutions of (1.1) will be denoted by $\psi_{m,n}(x)$, $n = m, m + 1, \cdots$. The eigenfunction $\psi_{m,n}(x)$ is an even or odd function of x according as $n - m$ is an even or odd integer, and $\psi_{m,n}$ has precisely $n - m$ zeros in the open interval $|x| < 1$.

The $\psi_{m,n}(x)$ are known as prolate spheroidal wave functions of order m. They have been extensively studied in the literature. Many details of their properties can be found in the books by Meixner and Schafke [1], Flammer [2], and Stratton, Morse, Chu, Little and Corbato [3]. The first two of these contain extensive bibliographies of the pertinent periodical literature prior to 1957.

Equation (1.1) results when the three-dimensional scalar wave equation is separated in a prolate spheroidal coordinate system. The $\psi_{m,n}$ arise in this connection in most of the physical applications cited in [1] and [2]. These deal with problems in acoustic or electromagnetic propagation involving ellipsoidal structures or with the quantum mechanical two-center problem.

Recently, however, the functions of zero order, $\psi_{0,n}(x)$, have come to play an important role in a wide variety of other applied problems from quite another connection. Central to these applications is the fact that the $\psi_{0,n}$ are the eigenfunctions of the finite Fourier transform operator, i.e., that they satisfy the integral equation

$$\alpha_n \psi_{o,n}(x) = \int_{-1}^{1} e^{icxy}\, \psi_{o,n}(y)\, dy \tag{1.2}$$

whose first iterate

$$\lambda_n \psi_{o,n}(x) = \int_{-1}^{1} \frac{\sin c(x - y)}{\pi(x - y)}\, \psi_{o,n}(y)\, dy, \qquad \lambda_n = \frac{c}{2\pi}\, |\alpha_n|^2 \tag{1.3}$$

also frequently occurs. Physical applications of this consequence are to such diverse fields as stochastic processes [4], the determination of laser modes [5], the uncertainty principle [10], the statistical theory of energy levels of complex systems [6], [7], antenna theory [8], and a variety of considerations of fundamental importance in communication theory [9]–[13]. New mathematical properties of the $\psi_{0,n}$ have come to light from some of these researches [9].

In many of these newer applications, the behavior of the $\psi_{0,n}(x)$ and the λ_n of (1.3) for large values of c are of considerable interest. A number of treatments of the asymptotics of prolate spheroidal wave functions for large c exist in the literature [1], [2], [14], [15] These do not suffice, however, to allow detailed determination of the $\psi_{m,n}(x)$ for all values of x, nor do they discuss the behavior of λ_n. Indeed, except for [15], these expansions give details about the prolate spheroidal functions (for fixed n and m) in an interval about $x = 0$ whose width unfortunately vanishes with increasing c like $c^{-\frac{1}{2}}$. The discussion in [15] provides information in the interval $|x| < 1 - 1/c$ when $n \ll c$.

In part II of this paper we develop expressions for $\psi_{m,n}(x)$ for fixed m and n valid for large values of c for all real values of x. In part III, formulae for n and c both large are obtained for all real values of x. In part IV the earlier results are applied to determine asymptotic expressions for the eigenvalues λ_n of (1.3) both for large and small c for fixed n and for n and c large together.

The results of parts II–IV are summarized in the next three sections of this introduction.

1.2 $\psi_{m,n}(x)$ *for Large c and Fixed m and n.* For fixed m and n and large c a solution of (1.1) is given by

$$\bar{\psi}_{m,n}(x) = \begin{cases} \psi^1_{m,n}(x), & 0 \leqq x \leqq c^{-\frac{1}{2}} \\ k_2\psi^2_{m,n}(x), & c^{-\frac{1}{2}} \leqq x \leqq 1 - c^{-1} \\ k_3\psi^3_{m,n}(x), & 1 - c^{-1} \leqq x \leqq 1 \\ k_4\psi^4_{m,n}(x), & 1 \leqq x \leqq 1 + c^{-1} \\ k_5\psi^5_{m,n}(x), & 1 + c^{-1} \leqq x \end{cases} \tag{1.4}$$

and the corresponding eigenvalue is given by

$$\frac{\chi_{m,n}}{2c} = n - m + \frac{1}{2} + \sum_{j=1} \left(\frac{1}{2c}\right)^j a_j \,. \tag{1.5}$$

Here

$$\psi^1_{m,n}(x) = (1 - x^2)^{m/2}\left[D_l(x\sqrt{2c}) + \sum_{j=1}\left(\frac{1}{2c}\right)^j \sum_{k=-2j}^{2j} A_k{}^j D_{l+2k}(x\sqrt{2c})\right] \tag{1.6}$$

$$\psi^2_{m,n}(x) = \frac{e^{cy}(1 - y)^{l/2}}{y^{\frac{1}{2}}(1 + y)^{(l+1)/2}}\left[1 + \sum_{j=1}\left(\frac{1}{2c}\right)^j \sum_{k=1}^{j}\left(\frac{D^j_k}{y^k} + \frac{E^j_k}{(1 + y)^k} + \frac{F^j_k}{(1 - y)^k}\right)\right] \tag{1.7}$$

$$\psi^3_{m,n}(x) = I_m(cy) + \sum_{j=1}\left(\frac{1}{2c}\right)^j \sum_{k=1}^{j} B^j_k(cy)^k I_{m+k}(cy) \tag{1.8}$$

$$\psi^4_{m,n}(x) = \epsilon_m \left[J_m(cz) + \sum_{j=1} \left(\frac{1}{2c}\right)^j \sum_{k=1}^{j} B^j_k(-cz)^k J_{m+k}(cz) \right] \tag{1.9}$$

$$\psi^5_{m,n}(x) = \operatorname{Re} \frac{e^{i(cz - m[\pi/2])}(1 - iz)^{l/2}}{(iz)^{\frac{1}{2}}(1 + iz)^{(l+1)/2}} \left[1 + \sum_{j=1} \left(\frac{1}{2c}\right)^j \sum_{k=1}^{j} \left(\frac{D^j_k}{(iz)^k} + \frac{E^j_k}{(1 + iz)^k} + \frac{F^j_k}{(1 - iz)^k} \right) \right] \tag{1.10}$$

where we have adopted the abbreviations $l = n - m$, $y = \sqrt{1 - x^2}$, $z = \sqrt{x^2 - 1}$ and $\epsilon_m = 1$ if m is odd while $\epsilon_m = (-1)^{m/2}$ if m is even

In the foregoing $D_\nu(x)$ is the Weber function of order ν (see [16], Vol. II, Chapter VIII), $J_m(x)$ is the usual Bessel function and $I_m(x)$ is the modified Bessel function. The quantities a_j, A^j_k, B^j_k, D^j_k, E^j_k, F^j_k are all polynomials in l and m. The a_j and A^j_k are given by the recurrence scheme (2.10)–(2.11), the B^j_k are given by the recurrence (2 54)–(2 55), the D^j_k, E^j_k, and F^j_k are determined by (2.32)–(2.34). The first few of each of these series of polynomials are given in Tables I, II, and III

The connecting constants of (1.4) are given by

$$k_2 = e^{-c} c^{l/2} 2^{(3l+1)/2} P(c) \tag{1.11}$$

$$k_3 = k_4 = e^{-c} c^{(l+1)/2} 2^{(3l+2)/2} \sqrt{\pi}\, P(c) Q(c) \tag{1.12}$$

$$k_5 = e^{-c} c^{l/2} 2^{(3l+3)/2} P(c). \tag{1.13}$$

TABLE I

$$\begin{aligned}
2a_0 &= 2l + 1 \\
-2^2 a_1 &= 2l^2 + 2l + 3 - 4m^2 \\
-2^3 a_2 &= (2l + 1)(l^2 + l + 3 - 8m^2) \\
-2^4 a_3 &= 5l^4 + 10l^3 + 40l^2 + 35l + 15 - 48m^2(2l^2 + 2l + 1) \\
-2^7 a_4 &= 66l^5 + 165l^4 + 962l^3 + 1278l^2 + 1321l + 453 \\
&\quad - m^2(2368l^3 + 3552l^2 + 4448l + 1632) + m^4(256l + 128)
\end{aligned}$$

$$\begin{aligned}
j!\,2^{4j} A^j_{2j} &= (-1)^j, && j = 0, 1, \cdots \\
(j-1)!\,2^{4j-3} A^j_{2j-1} &= (-1)^{j-1} m, && j = 1, 2, \cdots \\
(j-2)!\,2^{4j-3} A^j_{2j-2} &= (-1)^{j-1}(2l + 4j - 3 - 4m^2), && j = 2, 3, \cdots \\
(j-2)!\,2^{4j-3} A^j_{-2j+2} &= (2l - 4j + 5 + 4m^2) l(l-1) \quad (l - 4j + 5), && j = 2, 3, \cdots \\
(j-1)!\,2^{4j-3} A^j_{-2j+1} &= ml(l-1) \quad (l - 4j + 3), && j = 1, 2, \cdots \\
j!\,2^{4j} A^{j2}_{-j} &= l(l-1) \quad (l - 4j + 1), && j = 1, 2, \cdots
\end{aligned}$$

$$\begin{aligned}
32A^2_1 &= -m(l^2 - 25l - 36) \\
32A^2_{-1} &= m(l^2 - 27l - 10)l(l - 1) \\
3!\,2^9 A^3_3 &= m(3l^2 - 243l - 616) + 64m^3 \\
2^{13} A^3_2 &= -l^4 + 6l^3 - 715l^2 - 3514l - 4832 + m^2(3072l + 6656) \\
2^6 A^3_1 &= m(-6l^3 + 99l^2 + 275l + 252) + m^3(-4l^2 + 4l - 32) \\
2^6 A^3_{-1} &= [m(6l^3 + 117l^2 - 59l + 82) + m^3(-4l^2 - 12l - 40)]l(l - 1) \\
2^{13} A^3_{-2} &= [l^4 + 10l^3 + 739l^2 - 2062l + 2040 + m^2(3072l - 3584)]l(l - 1)(l - 2)(l - 3) \\
3!\,2^9 A^3_{-3} &= [m(3l^2 + 249l - 370) + 64m^3]l(l - 1)(l - 2)(l - 3)(l - 4)(l - 5)
\end{aligned}$$

TABLE II

$$
\begin{aligned}
B_1^1 &= -(2l+1) \\
2B_1^2 &= 2l^2+2l+3+4m \\
2B_2^2 &= 4l^2+4l+3 \\
4B_1^3 &= (2l+1)(l^2+l+3-8m^2) \\
4B_2^3 &= -(2l+1)(4l^2+4l+30+24m) \\
6B_3^3 &= -(2l+1)(4l^2+4l+15) \\
8B_1^4 &= 5l^4+10l^3+40l^2+35l+15-m^2(96l^2+96l+48) \\
8B_2^4 &= -4l^4-8l^3+30l^2+34l+75+m(48l^2+48l+168)+m^2(64l^2+64l+96) \\
4B_3^4 &= 8l^4+16l^3+170l^2+162l+105+m(80l^2+80l+60) \\
24B_4^4 &= 16l^4+32l^3+200l^2+184l+105
\end{aligned}
$$

TABLE III

$$
\begin{aligned}
j!2^{2j}D_j^j &= (1^2-4m^2)(3^2-4m^2)(5^2-4m^2)\cdots[(2j-1)^2-4m^2] \\
j!2^{2j}E_j^j &= (l+1)(l+2)\quad\cdot(l+2j) \\
j!2^{2j+1}D_j^{j+1} &= (2j+1)(2l+1)(1^2-4m^2)(3^2-4m^2)\cdots[(2j-1)^2-4m^2] \\
j!2^{2j+3}E_j^{j+1} &= -[l(l-4j-1)-2(j-1)-8m^2](l+1)(l+2) \\
&\quad \cdots(l+2j) \qquad\qquad j=1,2,3, \\
2^6D_1^3 &= [(l^2+l+1)(4m^2+135)-90](1-4m^2) \\
2^8E_1^3 &= -[14l^3-47l^2-7l-30-32m^2(7l-1)-32m^4](l+1)(l+2) \\
3!2^7D_1^4 &= [2265l^2+2265l+2070+m^2(156l^2+156l-268)+16m^4](2l+1)(1-4m^2) \\
3!2^7D_2^4 &= [455l^2+455l+140+4m^2(l^2+l+1)](1-4m^2)(9-4m^2) \\
4!2^9E_1^4 &= [l^6+l^5-1247l^4+3167l^3-686l^2+8700l+5184 \\
&\quad -m^2(16l^4-34336l^2+5520l+9952) \\
&\quad +m^4(192l^2+9792l-2560)+512m^6](l+1)(l+2) \\
4!2^9E_2^4 &= [l^4-110l^3+611l^2+554l+468-m^2(16l^2-1552l-480)+190m^4] \\
&\quad (l+1)(l+2)(l+3)(l+4) \\
F_k^j(l) &= (-1)^jE_k^j(-l-1), \qquad\qquad k=1,2,\cdot\cdot j;\ j=1,2,
\end{aligned}
$$

Here

$$P(c) = \left[1+\frac{1}{c}g_1^1+\frac{1}{c^2}g_2^1+\cdots\right]\Big/\left[1+\frac{1}{c}g_1^2+\frac{1}{c^2}g_2^2+\cdots\right] \tag{1.14}$$

$$Q(c) = \left[1+\frac{1}{c}h_1^2+\frac{1}{c^2}h_2^2+\cdots\right]\Big/\left[1+\frac{1}{c}h_1^3+\frac{1}{c^2}h_2^3+\cdots\right] \tag{1.15}$$

where the first few g's and h's are given on Table IV. It would appear that if $m = 0$, $Q(c) = 1$.

Normalization for the solution (1 4) is provided by

$$
\begin{aligned}
\frac{1}{N_{m,n}^2} &= \int_{-1}^{1}[\hat{\psi}_{m,n}(x)]^2\,dx \\
&= l!\sqrt{\frac{\pi}{c}}\Bigg[1+\frac{1}{2c}\{-m(2l+1)\} \\
&\qquad + \frac{1}{4c^2}\left\{\frac{l^4+2l^3+23l^2+22l+12}{2^7}\right. \\
&\qquad \left. + \frac{6m^2l(l+1)-6m(3l^2+3l+1)}{4}\right\}+\cdots\Bigg].
\end{aligned} \tag{1.16}
$$

TABLE IV

$$
\begin{aligned}
2^8 g_1^1 &= -\,l^4 - 10l^3 - 35l^2 - 50l - 24 + m(-128l - 64)\\
3!\,2^{17} g_2^1 &= l^8 + 36l^7 + 546l^6 + 3000l^5 + 3249l^4 - 24876l^3 - 93076l^2 - 119280l - 51840\\
&\quad + m(1536l^5 + 19200l^4 + 43008l^3 - 157440l^2 - 238080l - 129024)\\
&\quad + m^2(98304l^2 + 196608l + 73728)\\
3!\,6!\,2^{21} g_3^1 &= -\,l^{12} - 78l^{11} - 2717l^{10} - 44250l^9 - 311703l^8 + 745686l^7 + 17356009l^6\\
&\quad + 53924730l^5 - 95330356l^4 - 915844008l^3 - 2161406592l^2 - 2235703680l\\
&\quad - 857226240\\
&\quad + m(-5760l^9 - 233280l^8 - 3340800l^7 - 8208000l^6 + 164154240l^5 + 494804160l^4\\
&\quad - 1083651840l^3 - 3778133760l^2 - 4962309120l - 2097930240)\\
&\quad + m^2(-2949120l^6 - 57507840l^5 - 130129920l^4 + 979845120l^3 + 4844298240l^2\\
&\quad + 4346265600l + 1831403520)\\
&\quad + m^3(-188743680l^3 - 849346560l^2 + 47185920l + 212336640)\\
2^8 g_1^2 &= -\,l^4 - 10l^3 + 13l^2 + 62l + 64 - 128m^2\\
3!\,2^{17} g_2^2 &= l^8 + 36l^7 - 30l^6 - 4104l^5 - 18351l^4 + 13140l^3 + 112268l^2 + 265680l + 165888\\
&\quad + m^2(1536l^4 + 15360l^3 + 29184l^2 - 537600l - 491520) + 98304m^4\\
6!\,2^{22} g_3^2 &= -\,7l^{12} + 414l^{11} - 13019l^{10} + 230650l^9 - 2227641l^8 + 17132442l^7 + 102646817l^6\\
&\quad + 247576710l^5 - 710920612l^4 + 1293312584l^3 + 554188896l^2 + 2047668480l\\
&\quad + 902430720 + m^2(-1920l^8 - 69120l^7 - 188160l^6 + 11566080l^5 + 123461760l^4\\
&\quad + 319080960l^3 - 2668730880l^2 - 5287680000l - 3212574720) + m^4(-1474560l^4\\
&\quad - 14745600l^3 - 75202560l^2 + 940769280l + 975175680) - 62914560m^6\\
2^4 h_1^3 &= 6l + 3 + m(16l + 8) + m^2(8l + 4)\\
2^{10} h_2^3 &= 324l^2 + 324l + 171 + m(1152l^2 + 1152l + 480) + m^2(1248l^2 + 1248l + 328)\\
&\quad + m^3(512l^2 + 512l + 128) + m^4(64l^2 + 64l + 48)\\
2^{14}3^2 h_3^3 &= 4946l^3 + 74196l^2 + 99414l + 37341 + m(201792l^3 + 302688l^2 + 302928l\\
&\quad + 101016) + m^2(284768l^3 + 427152l^2 + 261592l + 59604) + m^3(185856l^3\\
&\quad + 278784l^2 + 130944l + 19008) + m^4(60032l^3 + 90048l^2 + 53536l + 11760)\\
&\quad + m^5(9216l^3 + 13824l^2 + 11520l + 3456) + m^6(512l^3 + 768l^2 + 2176l + 960)\\
2^4 h_1^2 &= 6l + 3 + m^2(8l + 4)\\
2^{10} h_2^2 &= 324l^2 + 324l + 171 + m^2(736l^2 + 736l + 200) + m^4(64l^2 + 64l + 48)\\
2^{14}3^2 h_3^2 &= 49464l^3 + 74196l^2 + 99414l + 37341 + m^2(146528l^3 + 219792l^2 + 130264l\\
&\quad + 28500) + m^4(23168l^3 + 34752l^2 + 25888l + 7152) + m^6(512l^3 + 768l^2\\
&\quad + 2176l + 960)
\end{aligned}
$$

The connection with Meixner's prolate function [1] is given by

$$ps_n^m(x, -c^2) = \left[\frac{2(n-m)!}{(2n+1)(n+m)!}\right]^{1/2} N_{m,n}\,\hat{\psi}_{m,n}(x), \tag{1.17}$$

and Flammer's angular function $S_{m,n}(c, x)$ [2] is related by

$$\hat{\psi}_{m,n}(x) = C_{m,n} S_{m,n}(c, x) \tag{1.18}$$

where if l is even

$$C_{m,n} = \frac{[\frac{1}{2}(n+m)]!\,2^{(n+m)/2}}{(n+m)!} \sum_{j=0} \left(\frac{1}{2c}\right)^j \sum_{k=-2j}^{2j} (-1)^k A_k^j \frac{(\frac{1}{2}l)!\,(l+2k)!}{(\frac{1}{2}l+k)!\,2^k} \tag{1.19}$$

whereas if l is odd

$$
\begin{aligned}
C_{m,n} = {}& \frac{[\frac{1}{2}(n+m+1)]!\,2^{(n+m)/2}}{(n+m+1)!}\,2\sqrt{c}\\
&\cdot \sum_{j=0} \left(\frac{1}{2c}\right)^j \sum_{k=-2j}^{2j} (-1)^k A_k^j \frac{[\frac{1}{2}(l-1)]!\,(l+2k)!}{[\frac{1}{2}(l-1)+k]!\,2^k}.
\end{aligned}
\tag{1.20}
$$

Flammer's radial function $R_{m,n}^{(1)}(c, x)$ [2] is connected with (1 4) by

$$R_{m,n}^{(1)}(c, x) = \frac{1}{ck_5} \hat{\psi}_{m,n}(x) \tag{1.21}$$

1.3 $\psi_{m,n}(x)$ *for Large c and n with m Fixed.* When n and c are both large, the eigenvalue of (1 1) can be written

$$\chi_{m,n} = c^2 + 2\delta c + \sum_{j=0} \frac{b_j}{c^j} \tag{1.22}$$

and the corresponding solution is

$$\hat{\hat{\psi}}_{m,n}(x) = \begin{cases} \psi^6_{m,n}(x), & 0 \leqq x \leqq 1 - c^{-\frac{1}{2}} \\ k_7\psi^7_{m,n}(x), & 1 - c^{-\frac{1}{2}} \leqq x \leqq 1 \\ k_8\psi^8_{m,n}(x), & 1 \leqq x \leqq 1 + c^{-\frac{1}{2}} \\ k_9\psi^9_{m,n}(x), & 1 + c^{-\frac{1}{2}} \leqq x. \end{cases} \tag{1.23}$$

Here

$$\psi^6_{m,n}(x) = \text{Re}\, \frac{\exp\left\{i\left(cx + \frac{\delta}{2}\log\frac{1+x}{1-x} - l\frac{\pi}{2}\right)\right\}}{\sqrt{1-x^2}} \left[1 + \sum_{j=1}\frac{1}{c^j}\sum_{k=1}^{j}\left(\frac{M_k^j}{(1+x)^k} + \frac{N_k^j}{(1-x)^k}\right)\right], \tag{1.24}$$

$$\psi^7_{m,n}(x) = (1 - x^2)^{m/2}U_{m,n}(x), \qquad \psi^8_{m,n}(x) = (x^2 - 1)^{m/2}U_{m,n}(x) \tag{1.25, 26}$$

$$U_{m,n}(x) = e^{ic(1-x)}\left[\Phi(\beta, m + 1; \xi) + \sum_{j=1}\frac{1}{c^j} V_j(\xi)\right] \tag{1.27}$$

$$V_j(\xi) = \Phi(\beta, m + 1; \xi)\sum_{k=1}^{j} K_k^j \xi^k + \xi\Phi(\beta + 1, m + 3, \xi)\sum_{k=1}^{j} L_k^j \xi^k \tag{1.28}$$

where $\xi = -2ic(1 - x)$, $\beta = \frac{1}{2}(m + 1 - i\delta)$

$$\psi^9_{m,n}(x) = \text{Re}\, \frac{\exp\left\{i\left[cx + \frac{\delta}{2}\log\left(\frac{x+1}{x-1}\right) - (n+1)\frac{\pi}{2}\right]\right\}}{c\sqrt{x^2 - 1}} \left[1 + \sum_{j=1}\frac{1}{c^j}\sum_{k=1}^{j}\left(\frac{M_k^j}{(1+x)^k} + \frac{N_k^j}{(1-x)^k}\right)\right]. \tag{1.29}$$

In the foregoing, $\Phi(a, c, x)$ is the confluent hypergeometric function ([16], Vol. 1, Chapter 6). The quantity δ is a function of c, m, and n. It must be chosen along with a suitable integer q to satisfy

$$c + \delta\log(2\sqrt{c}) - \phi_m(\delta) = (n + 1)\tfrac{1}{2}\pi - (m + 1)\tfrac{1}{4}\pi + 2\pi q \tag{1.30}$$

and when used with (1.23) must yield a solution $\hat{\hat{\psi}}_{m,n}$ having precisely $n - m$ zeros in $|x| < 1$. Here the function $\phi_x(y)$ is defined by

$$\Gamma[\tfrac{1}{2}(1 + x + iy)] = R_x(y)e^{i\varphi_x(y)} \tag{1.31}$$

with $\phi_x(0) = 0$ More detail about the selection of δ is given in Section 3.7.

The quantities b_j, K_k^j, L_k^j, M_k^j and N_k^j are all polynomials in δ. The first three of these are determined by the recurrence scheme (3 24)–(3.28); the M's and N's are given by a modification of the recurrence (2.32)–(2 34) as discussed below Eq. (3 36). The first few of each of these series of polynomials are given in Tables V and VI

The connecting constants of (1 23) are given by

$$k_7 = k_8 = \frac{c^{(m+1)/2}R_m(\delta)e^{\delta(\pi/4)}}{2\Gamma(m+1)}\left[1 + O\left(\frac{1}{c}\right)\right] \tag{1.32}$$

$$k_9 = ce^{\delta(\pi/2)}\left[1 + O\left(\frac{1}{c}\right)\right] \tag{1.33}$$

with $R_m(\delta)$ defined as in (1.31).

The solution (1 23) is connected with Flammer's radial function by

$$\hat{\hat{\psi}}_{m,n}(x) = k_9 R_{m,n}^{(1)}(c, x) \tag{1.34}$$

1.4 *Asymptotics of* λ_n. Regarding the eigenvalue λ_n of (1 3) we show in part IV that for small c,

$$\lambda_n = \frac{2}{\pi}\left[\frac{2^{2n}(n!)^3}{(2n)!\,(2n+1)!}\right]^2 c^{2n+1}\exp\left\{-\frac{(2n+1)c^2}{(2n-1)^2(2n+3)^2}[1 + O(c^2)]\right\} \tag{1.35}$$

while for fixed n and large c

TABLE V

$$\begin{aligned}
2b_0 &= \delta^2 - 1 + m^2, \qquad 2^3 b_1 = -\delta^3 + \delta - m^2\delta \\
2^6 b_2 &= 5\delta^4 - 10\delta^2 + 1 + m^2(6\delta^2 - 2) + m^4 \\
2^9 b_3 &= -33\delta^5 + 114\delta^3 - 37\delta + m^2(-46\delta^3 + 50\delta) - 13m^4\delta \\
2^{10} b_4 &= 63\delta^6 - 340\delta^4 + 239\delta^2 - 14 + m^2(100\delta^4 - 230\delta^2 + 30) \\
&\quad + m^4(39\delta^2 - 18) + 2m^6 \\
8(m+1)K_1^1 &= i[-\delta^2 + (m+1)^2], \qquad 16(m+1)^2(m+2)L_1^1 = \delta^3 + \delta(m+1)^2 \\
2^7(m+1)K_2^2 &= 5\delta^2 - 3 + m(3\delta^2 - 7) - 5m^2 - m^3 \\
2^5(m+1)K_1^2 &= i[\delta^3 - \delta + m^2\delta], \qquad 2^7(m+1)^2 L_2^2 = i[\delta^3 + \delta + 2m\delta + m^2\delta] \\
2^7(m+1)^2(m+2)L_1^2 &= -3\delta^4 - 2\delta^2 + 1 + m(-\delta^4 - 7\delta^2 + 2) - 6m^2\delta^2 + m^3(-\delta^2 - 2) \\
&\quad - m^4 \\
3!2^9(m+1)K_3^3 &= i[-\delta^4 + 32\delta^2 - 15 + m(30\delta^2 - 38) + m^2(6\delta^2 - 32) - 10m^3 \\
&\quad - m^4 \\
3!2^8(m+1)K_2^3 &= -\delta^5 - 26\delta^3 + 23\delta + m(-11\delta^3 + 13\delta) + m^2(-\delta^3 - 19\delta) - 9m^3 \\
2^8(m+1)K_1^3 &= i[-5\delta^4 + 10\delta^2 - 1 + m^2(-6\delta^2 + 2) - m^4] \\
3!2^{10}(m+1)^2(m+2)L_3^3 &= \delta^5 - 22\delta^3 - 23 + m(-16\delta^3 - 64\delta) + m^2(-2\delta^3 - 62\delta) \\
&\quad - 24m^3\delta - 3m^4\delta \\
3!2^9(m+1)^2(m+2)L_2^3 &= i[-\delta^6 - 29\delta^4 - 19\delta^2 + 9 + m(-17\delta^4 - 68\delta^2 + 21) \\
&\quad + m^2(-4\delta^4 - 70\delta^2 + 6) + m^3(-24\delta^2 - 18) + m^4(-3\delta^2 - 15) \\
&\quad - 3m^5 \\
3!2^8(m+1)^2(m+2)L_1^3 &= 23\delta^5 - 2\delta^3 - 25\delta + m(9\delta^5 + 52\delta^3 - 53\delta) + m^2(\delta^5 + 55\delta^3 - 18\delta) \\
&\quad + m^3(14\delta^3 + 26\delta) + m^4(\delta^3 + 19\delta) + 3m^5\delta
\end{aligned}$$

TABLE VI

$$
\begin{aligned}
N_k^j &= (M_k^j \text{ complex conjugate}) \\
2^3M_1^1 &= -2\delta + i(\delta^2 + m^2 - 1) \\
2^7M_1^2 &= \delta^4 + 10\delta^2 + 1 + m^2(2\delta^2 - 2) + m^4 + i\delta[-4\delta^2 + 4 - 4m^2] \\
2^7M_2^2 &= -\delta^4 + 22\delta^2 - 9 + m^2(-2\delta^2 + 10) - m^4 + i\delta[-8\delta^2 + 24 - 8m^2] \\
2^{10}M_1^3 &= -7\delta^5 - 62\delta^3 + 9\delta + m^2(-14\delta^3 - 2\delta) - 7m^4\delta + i[20\delta^4 - 40\delta^2 + 4 \\
&\quad + m^2(24\delta^2 - 8) + 4m^4] \\
2^{11}M_2^3 &= 2\delta^5 - 188\delta^3 + 66\delta + m^2(4\delta^3 - 68\delta) + 2m^4\delta + i[\delta^6 + 57\delta^4 - 209\delta^2 - 9 \\
&\quad + m^2(3\delta^4 + 46\delta^2 + 19) + m^4(3\delta^2 - 11)] \\
3!2^9M_3^3 &= 18\delta^5 - 444\delta^3 + 690\delta + m^2(36\delta^3 - 324\delta) + 18m^4\delta + i[-\delta^6 + 127\delta^4 - 799\delta^2 \\
&\quad + 225 + m^2(-3\delta^4 + 162\delta^2 - 259) + m^4(-3\delta^2 + 35)] \\
4!2^{13}M_1^4 &= \delta^8 + 1252\delta^6 + 11030\delta^4 - 7804\delta^2 + 81 + m^2(4\delta^6 + 2612\delta^4 + 2492\delta^2 - 52) \\
&\quad + m^4(6\delta^4 + 1468\delta^2 - 138) + m^6(4\delta^2 + 108) + m^8 + i[-3168\delta^5 + 10944\delta^3 \\
&\quad - 3552\delta + m^2(-4416\delta^3 + 4800\delta) - 1248m^4\delta] \\
4!2^{13}M_2^4 &= \delta^8 + 100\delta^6 + 16214\delta^4 - 13948\delta^2 + 657 + m^2(4\delta^6 + 116\delta^4 + 8636\delta^2 - 1396) \\
&\quad + m^4(6\delta^4 - 68\delta^2 + 822) + m^6(4\delta^2 - 84) + m^8 + i[-104\delta^7 - 3976\delta^5 \\
&\quad + 23080\delta^3 - 3768 + m^2(-312\delta^5 - 3696\delta^3 + 3592\delta) + m^4(-312\delta^3 + 280\delta) \\
&\quad - 104m^6\delta] \\
4!2^{11}M_3^4 &= -\delta^8 - 124\delta^6 + 5290\delta^4 - 8636\delta^2 - 225 + m^2(-4\delta^6 - 212\delta^4 + 3316\delta^2 + 484) \\
&\quad + m^4(-6\delta^4 - 52\delta^2 - 294) + m^6(-4\delta^2 + 36) - m^8 + i[-4\delta^7 - 1316\delta^5 \\
&\quad + 10052\delta^3 - 2460\delta + m^2(-12\delta^5 - 1464\delta^3 + 2612\delta) + m^4(-12\delta^3 - 148\delta) - 4m^6\delta] \\
4!2^{12}M_4^4 &= \delta^8 - 428\delta^6 + 13238\delta^4 - 49036\delta^2 + 11025 + m^2(4\delta^6 - 940\delta^4 + 13532\delta^2 - 12916) \\
&\quad + m^4(6\delta^4 - 596\delta^2 + 1974) + m^6(4\delta^2 - 84) + m^8 + i[32\delta^7 - 3104\delta^5 + 33632\delta^3 \\
&\quad - 36960\delta + m^2(96\delta^5 - 4800\delta^3 + 20192\delta) + m^4(96\delta^3 - 1696\delta) + 32m^6\delta]
\end{aligned}
$$

$$1 - \lambda_n = \frac{4\sqrt{\pi}2^{3n}c^{n+\frac{1}{2}}e^{-2c}}{n!}\left[1 - \frac{6n^2 - 2n + 3}{32c} + O(c^{-2})\right]. \quad (1.36)$$

The lead term here has been given previously by Fuchs [17].

Let δ be the root of smallest absolute value of

$$c + \delta \log(2\sqrt{c}) - \phi_0(\delta) = \tfrac{1}{2}\pi n - \tfrac{1}{4}\pi. \quad (1.37)$$

Then for large n and c, we have the approximation

$$\lambda_n \approx (1 + e^{\pi\delta})^{-1} \quad (1.38)$$

(Some numerical examples of the remarkable accuracy of this formula are given in the next section.) Finally we have the interesting limiting result that if b is fixed and

$$n = [(2/\pi)(c + b \log(2\sqrt{c}))] \quad (1.39)$$

where the brackets denote "largest integer in", then

$$\lim_{c\to\infty}\lambda_n = (1 + e^{\pi b})^{-1} \quad (1.40)$$

The case where $c = O((n/\log n)^{\frac{1}{2}})$ has been treated by Widom [19] who shows that then $\lambda_n \sim 2\pi(c/4)^{2n+1}/(n!)^2$.

1.5 *Some Numerical Examples.* Figure 1 shows the behavior of $\psi_{0,8}$ for $x > 0$ for $c = 1, 10, 20, 40, 80$. The function has been normalized according to Flammer's convention so that $\psi_{0,8}(0) = \frac{35}{128}$

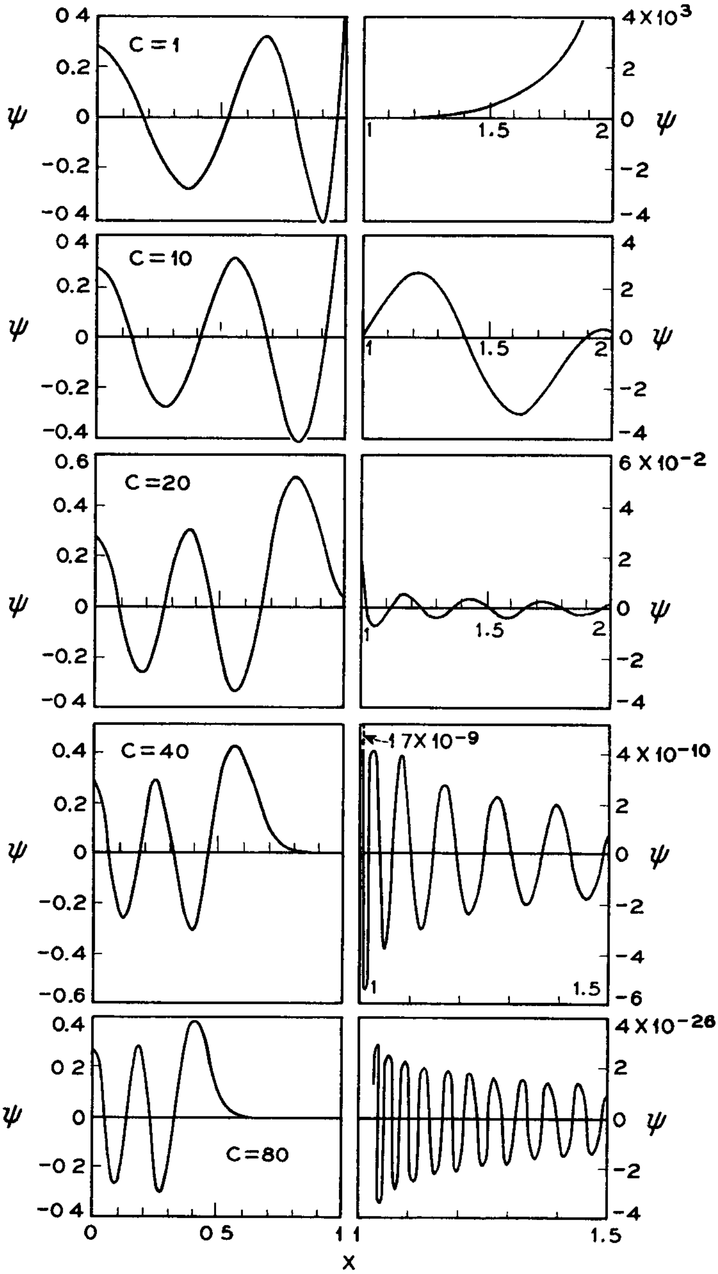

FIG 1 $\psi_{0,8}(x)$ for $c = 1, 10, 20, 40, 80$. Note change in vertical scales for $x > 1$ for each c and in horizontal scales for $x > 1$ for $c = 40, 80$.

For $c \ll (n - m)\pi/2$, $\psi_{m,n}$ is concentrated outside $|x| \leqq 1$. The zeros of ψ in $|x| \leqq 1$ are then distributed like those of $P_n^m(x)$, the associated Legendre function. When $c \gg (n - m)\pi/2$, $\psi_{m,n}$ is concentrated in $|x| \leqq 1$. Its $n - m$ zeros in $|x| \leqq 1$ are crowded in around the origin. For $|x| > 1$, $\psi_{m,n}$ oscillates rapidly and is small. These qualitative features can be seen clearly from the lead terms of (1.6)–(1 10)

When $c \approx (n - m)\pi/2$ and c is large, $\psi_{m,n}$ behaves like

$$(1 - x^2)^{-\frac{1}{2}} \begin{matrix}\cos\\ \sin\end{matrix} \left(cx + \frac{\delta}{2} \log \frac{1 + x}{1 - x}\right)$$

for $|x| < 1 - c^{-\frac{1}{2}}$ and for large x falls off like $1/x$ and oscillates with angular frequency c. These features can be seen from the lead terms of (1 24) and (1.29). The behavior near $x = 1$ is more difficult to describe (1.25)–(1 28).

Programs for the IBM 7090 have been written for the functions and constants of Eqs. (1.6)–(1.20) for the case $m = 0$. A program for computing $S_{0n}(c, x)$ and $R_{0n}^{(1)}(c, x)$ by means of Legendre polynomial and spherical Bessel function expansions as described in [2] Sections 3.1, 3.14, 3 15, and 4 1 has also been produced Tables VII, VIII, and IX illustrate the accuracy of the asymptotic formulae developed here The values in italics in the regions defined on the right of (1.4). The asterisks on Table IX indicate values of x for which the Legendre and Bessel expansions apparently do not converge.

A program was also developed to find the root of (1.37) and to compute the right of (1.24) Table X compares (1.24) with exact values.

TABLE VII

$n = 0, \quad c = 5, \quad m = 0$

x	ψ Exact	$k_1\psi_1$	$k_2\psi_2$	$k_3\psi_3$	$k_4\psi_4$	$k_5\psi_5$
1	979200-0	*979201-0*	982060-0			
2	918874-0	*918876-0*	921574-0			
3	824959-0	*824962-0*	827286-0			
4	706453-0	*706455-0*	708476-0			
5	574224-0	*.574252-0*	575927-0			
6	439613-0	*.439665-0*	440878-0	441537-0		
7	313042-0	313104-0	*314133-0*	.314160-0		
8	202850-0	199888-0	*203822-0*	*203516-0*		
9	114494-0	112213-0	115540-0	*.114851-0*		
1 0	502298-1	.477134-1		*503853-1*	*503853-1*	
1 1	926680-2				*927107-2*	.922680-2
1 2	− 116568-1				− *117108-1*	− 117065-1
1 3	− 174198-1				− 174587-1	− *.174712-1*
1 4	−.135429-1				− 134456-1	− *135788-1*
1 5	− 526640-2					− *528053-2*
1 6	317839-2					*318676-2*
1 7	900601-2					*.903072-2*
1 8	109912-1					*110216-1*
1 9	928790-2					*931279-2*
2 0	503154-2					*504573-2*

TABLE VIII

$n = 4, \quad c = 10, \quad m = 0$

x	ψ Exact	$k_1\psi_1$	$k_2\psi_2$	$k_3\psi_3$	$k_4\psi_4$	$k_5\psi_5$	$k_6\psi_6$
1	239277–0	*.239286–0*					*239011–0*
2	− 739182–1	− *739087–1*	− 748163–1				− *746155–1*
3	− 347042–0	− *347069–0*	−.342517–0				− *347587–0*
4	− 386654–0	− *386715–0*	− 386189–0				− *386105–0*
5	− 151237–0	− *151234–0*	− 149464–0				−.149155–0
6	227172–0	227352–0	*230211–0*				230185–0
7	536608–0	541246–0	*538739–0*				.539381–0
8	623349–0	622695–0	*617992–0*	.635883–0			.624301–0
9	478826–0	473955–0	*.481653–0*	*487719–0*			.398753–0
1 0	223313–0	202390–0		*.227040–0*	*227040–0*		
1.1	103927–1				*.101958–1*	*104520–1*	
1 2	− 727571–1				−.737972–1	−*.755837–1*	
1 3	− 421430–1					− *421372–1*	
1 4	221864–1					*.222889–1*	
1 5	488366–1					*502401–1*	
1 6	229148–1					*229849–1*	
1 7	− 200298–1					−*.201089–1*	
1 8	− 372073–1					− *379807–1*	
1 9	− 171842–1					−*.172361–1*	
2.0	160907–1					*161506–1*	

Figure 2 shows the dependence on c of the eigenvalues λ_n of (1 3). Exact values of λ_n were obtained from (4.1) using the spherical Bessel function expansion of [2] Section 4 1 to compute $R_{0n}^{(1)}(c, 1)$. Table XI compares the approximation (1 37)–(1 38) with these exact values. For λ in the range $.1 < \lambda < .95$ the approximation is already good to 1% for n as small as 3

1.6 *Discussion.* In many ways the material presented here is far from complete From the purely mathematical point of view, our derivations must be regarded as heuristic The infinite series in the formulae (1.6)–(1.10) of Section 1 2 and (1 24)–(1.29) of Section 1.3 appear to be divergent. We suspect that they are all true asymptotic series in the sense that if truncating one after N terms yields ψ^N, then $\lim_{c\to\infty} c^{N/3}(\psi - \psi^N) = 0$ where ψ is the properly normalized exact solution of (1.1). We have not, however, established this fact.

In the matching of solutions presented here there is also much to be desired. The nonlinear recurrences encountered for the various coefficients in the expansions greatly complicate matters and prevent writing general expressions We must be content with showing that a few explicitly computed terms match Indeed, were it not for the ALPAK program [18] even this modest goal would not have been fulfilled, the algebra becomes so burdensome. Tables I–VI were computed by this program

In addition to the foregoing shortcomings, it must be added that the choice of δ described in part III is particularly troublesome

Despite these difficulties, however, there can be little doubt that the ex-

TABLE IX

$n = 4, \quad c = 30, \quad m = 0$

x	ψ Exact	$k_1\psi_1$	$k_2\psi_2$	$k_3\psi_3$	$k_4\psi_4$	$k_5\psi_5$
10	− 144564–1	− *144564–1*	− 144557–1			
20	− 392575–0	− *392575–0*	− 392541–0			
30	− 230669–1	− *230669–1*	− 230649–1			
40	450703–0	*450703–0*	*450664–0*			
50	429440–0	429440–0	*429403–0*			
60	180140–0	180140–0	*180125–0*			
70	370492–1	370492–1	*370460–1*			
80	327512–2	327511–2	*327484–2*	314643–2		
85	628793–3	628784–3	*628738–3*	.627696–3		
90	77006?–4	770282–4	*.770030–4*	770015–4		
95	*	430050–5	*428524–5*	*.428524–5*		
1 00	*	− 105885–6		*.112269–7*	*112269–7*	
1 05	*				*146396–8*	*146395–8*
1 10	*				− 129693–9	− *129702–9*
1.15	*				642373–9	*642057–9*
1.20	*				− 156789–8	− *151488–8*
1 30	*					− *781949–9*
1.40	*					*144356–8*

TABLE X

$n = 8, \quad m = 0, \quad c = 10, 20$

x	$c = 10$		$c = 20$	
	Exact	ψ^6	Exact	ψ^6
.05	231111–0	231021–0	177887–0	177531–0
.10	116808–0	116504–0	− 427151–1	− 436431–1
15	− 350653–1	− 355691–1	− 235749–0	− 236486–0
20	− 178271–0	− 178799–0	− 267050–0	− 266438–0
.25	− 268208–0	− 268473–0	− 112882–0	− 110591–0
30	− 275231–0	− 274949–0	123951–0	126681–0
35	− 194072–0	− 193104–0	283803–0	284722–0
40	− 465647–1	− 450301–1	257069–0	254605–0
45	123407–0	125092–0	560376–1	508860–1
.50	261736–0	262930–0	− 194419–0	− 199137–0
.55	319843–0	319847–0	− 337678–0	− 338115–0
.60	270264–0	268545–0	281998–0	− 276217–0
.65	117811–0	114214–0	− 510687–1	− 409143–1
70	− 970723–1	− 102176–0	238688–0	247810–0
.75	− 304332–0	− 309820–0	452400–0	453954–0

pressions given are the correct asymptotic ones, they are remarkedly internally consistent and check well numerically. We have accordingly deemed it best to present the material now in this unfinished state rather than postpone publication to some unforeseeable date when a more rigorous derivation may be possible

1.7 *Acknowledgment* The author is indebted to Mrs L. Needham for the ALPAK computation reporter here and to Mrs E Sonnenblick for the extensive IBM 7090 programming used to numerically verify the many formulae developed here Both sets of computations helped guide the author and brought to light several errors in his original theoretical treatment of the problem

II. Derivation of Asymptotics for Fixed m and n and Large c. 2.1 *Behavior Near the Origin.* For completeness we present here a short derivation of the standard asymptotic expansion of the $\psi_{m,n}(x)$ to be found in [1] and [2] The recurrence scheme (2 10)–(2 11) does not appear in the literature.

In (1 1), set $\psi(x) = (1 - x^2)^{m/2}u(x)$ and $x = t/\sqrt{2c}$ There results

$$\left(\mathcal{G} - \frac{1}{2c}\mathcal{K}\right)u(t) + \frac{\chi}{2c}u(t) = 0 \tag{2.1}$$

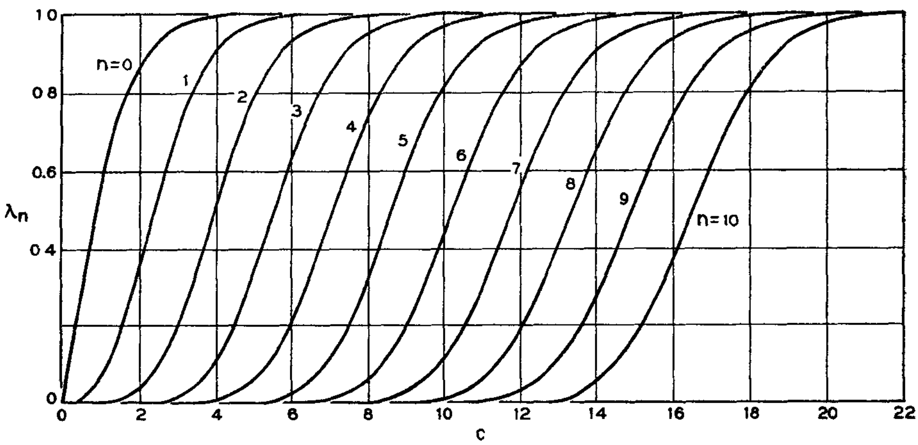

FIG 2 Curves of λ_n versus c for $n = 0$ to 10

where the operators $\mathcal{G}$ and $\mathcal{K}$ are given by

$$\mathcal{G} = \frac{d^2}{dt^2} - \frac{t^2}{4}, \qquad \mathcal{K} = t^2\frac{d^2}{dt^2} + 2(m+1)t\frac{d}{dt} + m(m+1). \tag{2.2}$$

We seek a bounded solution of (2.1) having $l = n - m$ zeros in $|t| < \sqrt{2c}$.

Now the Weber functions $D_j(t)$, $j = 0, 1, 2, \cdots$, (see [16], Vol. II. Chapter VIII) satisfy

$$\mathcal{G}D_j(t) + (j + \tfrac{1}{2})D_j(t) = 0 \tag{2.3}$$

and $D_j(t)$ has precisely j zeros. Equation (2.1) indicates that for fixed t and very large c we should have approximately $\psi_{m,n} = (1 - x^2)^{m/2}u_l(t)$ where $u_l(t) \approx D_l(t)$, $\chi_{n,m}/2c \approx l + \frac{1}{2}$. This suggests attempting a formal solution of (2.1) by the series

TABLE XI

c	exact λ	λ from (1 39)
	$n = 0$	
5	309690	.335527
1.0	.572582	599627
1.5	762590	773452
2.0	880560	881136
	$n = 3$	
3.5	0492455	0468574
4.0	110211	109229
5 0	.343562	.346547
6 0	646792	.649512
7.0	864567	.864608
8 0	960546	.959634
9 0	.990395	989777
	$n = 10$	
13 5	.0307501	0302790
14 5	.102185	.101969
15.5	.261187	.261725
18 0	.825431	.825589
19 5	.962447	962137
20 5	.988687	988456

$$u_l(t) = D_l(t) + \sum_{j=1}^{\infty} \left(\frac{1}{2c}\right)^j Q_j(t) \tag{2.4}$$

$$\frac{\chi_{n,m}}{2c} = l + \frac{1}{2} + \sum_{j=1}^{\infty} \left(\frac{1}{2c}\right)^j a_j \tag{2.5}$$

where the a's and Q's do not depend on c. Substituting these series into (2.1) and equating to zero the coefficients of distinct powers of c, we find the system of equations

$$\begin{aligned} \mathcal{G}D_l + (l + \tfrac{1}{2})D_l &= 0 \\ \mathcal{G}Q_j + (j + \tfrac{1}{2})Q_j &= \mathcal{K}Q_{j-1} - \textstyle\sum_{k=1}^{j} a_k Q_{j-k}, \quad j = 1, 2, \cdots . \end{aligned} \tag{2.6}$$

Here we define $Q_0 = D_l$.

Now it is easy to derive from the recurrence relations for Weber functions ([16], Vol II, p. 119) that

$$\mathcal{K}D_l = \textstyle\sum_{i=-2}^{2} \gamma_l^i D_{l+2i} \tag{2.7}$$

where

$$\begin{aligned} \gamma_l^{-2} &= \tfrac{1}{4}l(l-1)(l-2)(l-3), \qquad \gamma_l^{-1} = ml(l-1), \\ \gamma_l^0 &= m^2 - \tfrac{1}{4}(2l^2 + 2l + 3), \qquad \gamma_l^1 = -m, \qquad \gamma_l^2 = \tfrac{1}{4}. \end{aligned} \tag{2.8}$$

(The superscript on γ is not, of course, an exponent. We shall often use such superscripts.) From (2 7) and (2.3), it is seen that the solution of (2.6) is of the form

$$Q_j(t) = \sum_{k=-2j}^{2j} A_k^j D_{l+2k}(t) \tag{2.9}$$

On substituting this form into (2 6) and equating to zero the coefficient of D_p for each p, we obtain the recurrence relations*

$$a_j = \sum_{k=-2}^{2} A_{-k}^{j-1} \gamma_{l-2k}^{k}, \qquad j = 1, 2, \tag{2.10}$$

$$\begin{aligned} 2rA_r^j &= \sum_{k=1}^{j} a_k A_r^{j-k} - \sum_{k=-2}^{2} A_{r-k}^{j-1} \gamma_{l+2(r-k)}^{k}, \\ r &= -2j, -2j+1, \cdots, 2j; \qquad j = 1, 2, \cdots. \end{aligned} \tag{2.11}$$

Here we have adopted the conventions $a_0 = 0$, $A_k^0 = \delta_{0k}$ and if $j \neq 0$, $A_k^j = 0$ if $|k| > 2j$, or $k = 0$, or $k < -l/2$.

Combining results, we have a formal solution of (1.1)

$$\psi_{m,n}^1(x) = \left(1 - \frac{t^2}{2c}\right)^{m/2} \left[D_l(t) + \sum_{j=1}^{\infty} \left(\frac{1}{2c}\right)^j Q_j(t)\right] \tag{2.12}$$

$$\chi_{m,n} = (2l+1)c + a_1 + \sum_{j=1}^{\infty} \left(\frac{1}{2c}\right)^j a_{j+1} \tag{2.13}$$

where $l = n - m$, $t = x\sqrt{2c}$, Q_j is given by (2.9)) and the a's and A's by the recurrence (2 10)–(2.11) While tedious for hand calculations in literal terms, this recurrence is easily handled for numerical work by high speed computers. Expressions for the first few a's and A's are given in Table I.

The series (2.12) is ordered in powers of $(2c)^{-1}$ when expressed in the variable t. For fixed t, then, the first few terms can be expected to give a good approximation to ψ if c is made large enough. However, fixed t, say $t = k_1$, implies $x = k_1/\sqrt{2c}$, so that the x region for which (2.12) gives useful information about ψ vanishes with increasing c. When $t = x\sqrt{2c}$ is substituted in the right of (2.12), the sum is no longer ordered in powers of c^{-1}, for, since $D_{j+2}(x\sqrt{2c})/D_j(x\sqrt{2c}) = O(c)$, it is seen from (2.9) that $Q_{j+1}(x\sqrt{2c})/Q_j(x\sqrt{2c})$ is $O(c^2)$. For fixed x, the terms of (2.12) are of increasing order in c. While it is true, as proved in both [1] and [14] that the solution of (1 1) is given by

$$\psi_{m,n}(x) = (1 - x^2)^{m/2} D_{n-m}(x\sqrt{2c}) + O(1/c),$$

this statement is in a sense vacuous as a description of $\psi_{m,n}(x)$ Since for fixed x, $D_p(x\sqrt{2c})$ approaches zero exponentially, it says only that except for $x = 0$, $\psi_{m,n}$ approaches zero with increasing c at least as fast as $1/c$. For large c, except very near $x = 0$, ψ does not look like $(1 - x^2)^{m/2} D_{n-m}(x\sqrt{2c})$, other than in the weak sense that both are small.

The situation regarding $\chi_{m,n}$ is more satisfactory Sips [14] has shown that (2 13) is an asymptotic series, so that

* The procedure outlined here is a special case of a general perturbation scheme described in Appendix I of Ref 20

$$\lim_{c\to\infty} c^N \left(\chi_{n,m} - (2l+1)c - a_1 - \sum_{j=1}^{N} \left(\frac{1}{2c}\right)^j a_{j-1}\right) = 0.$$

2.2 *The Oscillatory Regions of* $\psi_{m,n}(x)$. In (1.1) substitute $\psi = (1 - x^2)^{-1/2}U$. There results

$$\ddot{U} + r(x)U = 0 \tag{2.14}$$

where

$$r(x) = \frac{1 - m^2 + (\chi - c^2x^2)(1 - x^2)}{(1 - x^2)^2} = c^2 \frac{(x^2 - x_0^2)(x^2 - x_1^2)}{(1 - x^2)^2} \tag{2.15}$$

where x_0^2 and x_1^2 are the roots of the quadratic equation

$$x^2 - (1 + \chi/c^2)x + (\chi + 1 - m^2)/c^2 = 0. \tag{2.16}$$

Now from (2.14) it is seen that U can have an oscillatory behavior only in regions where $r(x) > 0$. In intervals where $r(x) < 0$, U can have at most one zero. Since the factor $(1 - x^2)^{-1/2}$ relating ψ and U is monotone in every interval not including the points $x = \pm 1$, the oscillatory behavior of ψ in such regions is the same as that of U.

From (2.13), one has for large c, $\chi_{n,m} = (2l + 1)c + a_1 + O(1/c)$. Substituting this value for χ in (2.16) one finds

$$\begin{aligned} x_0 &= \pm\sqrt{\frac{2l+1}{c}}\left[1 + O\left(\frac{1}{c}\right)\right] \\ x_1 &= \pm\left(1 - \frac{1}{c^2}\frac{(1 - m^2)}{2}\right) + O\left(\frac{1}{c^3}\right). \end{aligned} \tag{2.17}$$

Examination of (2.15) then shows that for large c, the oscillations of ψ are confined to the regions $|x| < \sqrt{(2l+1)/c}$ and $|x| > 1 - (1 - m^2)/2c^2$.

Since $\psi_{m,n}(x)$ is either even or odd in x, in all that follows we shall restrict our attention to the region $x \geqq 0$. We shall find different asymptotic expressions for ψ in each of the regions defined on the right of (1.4). The separate consideration of regions near $x = 1$ is necessitated by the singular behavior of $r(x)$ at $x = 1$.

For $0 \leqq x \leqq c^{-1/3}$ the successive term of (2.12) are ordered in powers of $c^{-1/3}$, and for large enough c, the first few terms of this series give an accurate description of $\psi_{m,n}$.

2.3 *Solution for* $c^{-1/3} \leqq x \leqq 1 - c^{-1}$ It is now convenient to work with the independent variable $y = \sqrt{1 - x^2}$. Equation (1.1) becomes

$$(1 - y^2)\frac{d^2\psi}{dy^2} + \left(\frac{1}{y} - 2y\right)\frac{d\psi}{dy} + \left(\chi - c^2 + c^2y^2 - \frac{m^2}{y^2}\right)\psi = 0. \tag{2.18}$$

In this latter equation substitute

$$\psi = \frac{e^{cy}(1 - y)^{l/2}}{\sqrt{y}(1 + y)^{(l+1)/2}}\, v(y) \tag{2.19}$$

and the asymptotic series (2.13) for $\chi_{m,n}$. There results

$$(1 - y^2)\dot{v} + [2c(1 - y^2) - (2l + 1)]v + \left[\frac{\frac{1}{4} - m^2}{y^2} + \frac{(l + \frac{1}{2})^2 + \frac{3}{4} - (2l + 1)y}{(1 - y^2)} + a_1 + \sum_{j=1}^{\infty} \frac{a_{j+1}}{(2c)^j}\right] v = 0$$

or

$$\dot{v} + \frac{1}{2c}\mathcal{L}v + \frac{1}{1 - y^2}\sum_{2}^{\infty} \frac{1}{(2c)^j} a_j v = 0 \tag{2.20}$$

where the operator $\mathcal{L}$ is given by

$$\mathcal{L}v = \ddot{v} - \left(l + \frac{1}{2}\right)\left(\frac{1}{1 + y} + \frac{1}{1 - y}\right)\dot{v} + s(y)v \tag{2.21}$$

and

$$4s(y) = \frac{1 - 4m^2}{y^2} + \frac{(l + 1)(l + 2)}{(1 + y)^2} + \frac{l(l - 1)}{(1 - y)^2}. \tag{2.22}$$

Here we have made use of the explicit expression found earlier for a_1 as given in Table I

Substitute the expansion

$$v = 1 + \sum_{j=1} \frac{1}{(2c)^j} f_j(y) \tag{2.23}$$

into (2.20). Here the $f_j(y)$ are assumed independent of c. Equating to zero the coefficients of distinct powers of c, we obtain the system of equations

$$\begin{aligned} -\dot{f}_0 &= 0, \qquad -\dot{f}_1 = \mathcal{L}f_0 \\ -\dot{f}_j &= \mathcal{L}f_{j-1} + \frac{1}{1 - y^2}\sum_{i=2}^{j} a_i f_{j-i}, \qquad j = 2, 3, \cdots \end{aligned} \tag{2.24}$$

where we define $f_0 = 1$.

Equations (2.24) can now be integrated successively Each new integration introduces a constant of integration. These can be handled in a systematic way as follows. Let $f_0 = 1, f_1, f_2, \cdots$ be a particular solution of (2.24) Then we assert that

$$\begin{aligned} g_0 &= f_0 \\ g_j(y) &= f_j + \textstyle\sum_{r=1}^{j-1} k_r f_{j-r} + k_j = f_j + \sum_{r=1}^{j} k_r f_{j-r}, \quad j = 1, 2, \cdot \end{aligned}$$

which contains a new additive constant for each g, is also a solution for arbitrary values of the constants $k_1, k_2, \cdots$. For, we have

$$\begin{aligned} -\dot{g}_j &= -\dot{f}_j - \sum_{r=1}^{j} k_r \dot{f}_{j-r} \\ &= \mathcal{L}f_{j-1} + \frac{1}{1 - y^2}\sum_{i=2}^{j} a_i f_{j-i} + \sum_{r=1}^{j-1} k_r \left[\mathcal{L}f_{j-r-1} + \frac{1}{1 - y^2}\sum_{i=2}^{j-r} a_i f_{j-r-i}\right] \end{aligned}$$

$$= \mathfrak{L}\left(f_{j-1} + \sum_{r=1}^{j-1} k_r f_{j-1-r}\right) + \frac{1}{1-y^2}\sum_{i=2}^{j} a_i \left(f_{j-i} + \sum_{r=1}^{j-i} k_r f_{j-i-r}\right)$$

$$= \mathfrak{L}g_{j-1} + \frac{1}{1-y^2}\sum_{i=2}^{j} a_i\, g_{j-i}\,.$$

In view of this fact, we can now write in place of (2.23)

$$v = 1 + \frac{1}{2c}(f_1 + k_1) + \frac{1}{(2c)^2}(f_2 + k_1 f_1 + k_2) + \cdot$$

$$= 1 + \sum_{j=1}^{\infty} \frac{1}{(2c)^j}\left(f_j + \sum_{r=1}^{j} k_r f_{j-r}\right).$$

But now it is evident that the arbitrariness apparently introduced by these constants of integration into the determination of ψ is completely illusory. For, on writing

$$G(y) = \frac{e^{cy}(1-y)^{l/2}}{\sqrt{y}(1+y)^{(l+1)/2}} \tag{2.25}$$

equations (2.19) and the expression just found for v give us

$$\psi = G(y)\left[1 + \frac{1}{2c}(f_1 + k_1) + \frac{1}{(2c)^2}(f_2 + k_1 f_1 + k_2) + \cdots\right]$$

$$= G(y)\left[1 + \frac{1}{2c}f_1 + \frac{1}{(2c)^2}f_2 + \frac{1}{(2c)^2}f_3 + \quad\right]$$

$$+ \frac{k_1}{2c}G(y)\left[1 + \frac{1}{2c}f_1 + \frac{1}{(2c)^2}f_2 + \frac{1}{(2c)^2}f_3 + \cdots\right] + \cdots$$

$$= \left(1 + \sum_{1}^{\infty}\frac{k_i}{(2c)^i}\right)G(y)\sum_{0}^{\infty}\frac{1}{(2c)^j}f_j(y).$$

The choice of the k's formally affects only the scale of ψ. Accordingly, we henceforth write

$$\psi = G(y)\left[1 + \sum_{j=1}^{\infty}\frac{1}{(2c)^j}f_j(y)\right] \tag{2.26}$$

with G given by (2.25), and now turn our attention to obtaining a particular solution of (2 24).

The first of (2.24) is

$$-f_1 = \mathfrak{L}f_0 = s(y) = \frac{1}{4}\left[\frac{1-4m^2}{y^2} + \frac{(l+1)(l+2)}{(1+y)^2} + \frac{l(l-1)}{(1-y)^2}\right]$$

and we take

$$f_1(y) = \frac{1}{4}\left[\frac{1-4m^2}{y} + \frac{(l+1)(l+2)}{1+y} - \frac{l(l-1)}{1-y}\right], \tag{2.27}$$

Note that the apparent singularity at $y = 1$, i.e., $x = 0$, due to the term $l(l-1)/(1-y)$ is canceled by the factor $(1-y)^{l/2}$ in $G(y)$ in (2.26). Our solution so far is analytic at $y = 1$.

The next equation of (2.24) is

$$-\dot{f}_2 = \mathfrak{L}f_1 + \frac{a_2}{1-y^2}. \tag{2.28}$$

From (2.21), (2.22) and (2.27), it is easily seen that $\mathfrak{L}f_1$ on the right of (2.28) can be expanded in partial fractions in the form

$$\sum_{i=1}^{3}\left(\frac{D_i}{y^i} + \frac{E_i}{(1+y)^i} + \frac{F_i}{(1-y)^i}\right).$$

Actual computation shows that $D_1 = 0$ and that $E_1 = F_1 = (2l+1)(l^2+l+3-8m^2)/16$. But, by the recurrence (2.10), (2.11) one finds, as shown in Table I, $a_2/2 = -E_1$ and so the terms on the right of (2.28) that would integrate into logarithms all vanish One finds

$$\begin{aligned} f_2(y) = \frac{1}{2^5}\Bigg[&\frac{12(2l+1)(1-4m^2)}{y} + \frac{(1-4m^2)(9-4m^2)}{y^2} \\ &- \frac{(l+1)(l+2)[l(l-5)-8m^2]}{(1+y)} + \frac{(l+1)(l+2)(l+3)(l+4)}{(1+y)^2} \\ &- \frac{l(l-1)[(l+1)(l+6)-8m^2]}{1-y} + \frac{l(l-1)(l-2)(l-3)}{(1-y)^2}\Bigg]. \end{aligned} \tag{2.29}$$

Note that once again the coefficients of $1/(1-y)$ and $1/(1-y)^2$ are such that combined with the factor $(1-y)^{l/2}$ in G, ψ has no singularities at $y = 1$.

The foregoing examples suggest that a solution of (2.24) can be obtained in the form

$$f_j(y) = \sum_{i=1}^{j}\left(\frac{D_i^j}{y^i} + \frac{E_i^j}{(1+y)^i} + \frac{F_i^j}{(1-y)^i}\right). \tag{2.30}$$

Adopt the abbreviations $z_1 = y$, $z_2 = 1+y$, $z_3 = 1-y$. Now we can write for $i = 1, 2, 3$ and $k = 0, 1, 2, \cdots,$

$$\mathfrak{L}\frac{1}{z_i^k} = \sum_{l=1}^{k+2}\sum_{j=1}^{3}\frac{d_{kl}^{ij}}{z_j^l}, \qquad \frac{1}{1-y^2}\frac{1}{z_i^k} = \sum_{l=1}^{k+1}\sum_{j=1}^{3}\frac{u_{kl}^{ij}}{z_j^l}. \tag{2.31}$$

Insert these expansions along with (2.30) into the general equation of (2.24) By equating to zero the coefficients of distinct powers of $1/y$, $1/(1+y)$ and $1/(1-y)$, we obtain the recurrence

$$kD_j^k = e_{jk}^1, \qquad kE_j^k = e_{jk}^2, \qquad -kF_k^j = e_{jk}^3, \qquad k = 1, 2, \cdots, j, \quad j = 2, 3, \cdots \tag{2.32}$$

where we have written

$$e_{jk}^l = \sum_{i=k-1}^{j-1}(D_i^{j-1}d_{ik+1}^{1l} + E_i^{j-1}d_{ik+1}^{2l} + F_i^{j-1}d_{ik+1}^{3l}) + \sum_{r=2}^{j-k} a_r \sum_{i=k}^{j-r}(D_i^{j-r}u_{ik+1}^{1l} + E_i^{j-r}u_{ik+1}^{2l} + F_i^{j-r}u_{ik+1}^{3l}).$$

These permit one to obtain successively the D's, E's and F's assuming the a's known. In (2.32) terms preceded by $\sum_a^b$ are to be interpreted as zero if $b < a$, and the conventions $D_i^j = E_i^j = F_i^j = 0$ if $i < 1$ or $i > j$ or $j < 1$ have been adopted. The starting values can be taken from (2.27) or (2.29).

In order that (2.30) satisfy (2.24), in addition to (2.32) it is also necessary that the equations

$$e_{j0}^1 = 0, \qquad e_{j0}^2 = e_{j0}^3 = -a_j/2 \qquad j = 2, 3, \cdot \tag{2.33}$$

be satisfied. These insure that the logarithmic terms obtained on integrating (2.24) all vanish. Equation (2.32) together with the second or third equation of (2.33) can be considered as a recurrence to determine the D's, E's and F's and a's. That the remaining two equations of (2.33) will then be satisfied and that this recurrence will yield the same values of the a_j for all j as the recurrence (2.10)–(2.11) has not yet been shown. Computations by ALPAK for $j = 1$ to 5 have given the same a's and checked the consistency of (2.33).

The constants in the expansions (2.31) are readily worked out. They are

$$
\begin{aligned}
&d_{kj}^{21} = -\alpha k\delta_{j1} + \alpha\delta_{j2}, \qquad d_{kj}^{31} = \alpha k\delta_{j1} + \alpha\delta_{j2},\\
&d_{kj}^{12} = (-1)^k k(\beta - a_0)\delta_{j1} + (-1)^k\beta\delta_{j2}, \qquad d_{kj}^{32} = k(\beta - a_0)2^{-(k+1)}\delta_{j1} + \beta 2^{-k}\delta_{j2},\\
&d_{kj}^{13} = k(\gamma + a_0)\delta_{j1} + \gamma\delta_{j2}, \qquad d_{kj}^{23} = k(\gamma + a_0)2^{-(k+1)}\delta_{j1} + \gamma 2^{-k}\delta_{j2},\\
&d_{kj}^{11} = \begin{cases} ka_0[1 + (-1)^{k+j+1}] + [\gamma + \beta(-1)^{k+j}](k + 1 + j), & j \leqq k + 1\\ \alpha + k(k+1), & j = k + 2\\ 0, & j > k + 2 \end{cases}\\
&d_{kj}^{22} = \begin{cases} ka_0 2^{j-k-2} + (\alpha + \gamma 2^{j-k-2})(k + 1 - j), & j \leqq k + 1\\ (k+1) + ka_0 + \beta, & j = k + 2\\ 0, & j > k + 2 \end{cases}\\
&d_{kj}^{33} = \begin{cases} -ka_0 2^{j-k-2} + (\alpha + \beta 2^{j-k-2})(k + 1 - j), & j \leqq k + 1\\ k(k+1) - ka_0 + \gamma, & j = k + 2\\ 0, & j > k + 2 \end{cases}\\
&u_{kj}^{21} = u_{kj}^{31} = 0, \qquad u_{kj}^{12} = \tfrac{1}{2}(-1)^k\delta_{1j}, \qquad u_{kj}^{32} = 2^{-(k+1)}\delta_{1j},\\
&u_{kj}^{13} = \tfrac{1}{2}\delta_{1j}, \qquad u_{kj}^{23} = 2^{-(k+1)}\delta_{j1},\\
&u_{kj}^{11} = \begin{cases} \tfrac{1}{2}[1 + (-1)^{k+j}], & j \leqq k\\ 0, & j > k; \end{cases}\\
&u_{kj}^{22} = \begin{cases} 2^{j-k-2}, & j \leqq k + 1\\ 0, & j > k + 1 \end{cases}\\
&u_{kj}^{33} = \begin{cases} 2^{j-k-2}, & j \leqq k + 1\\ 0, & j > k + 1 \end{cases}
\end{aligned}
\tag{2.34}
$$

where δ_{ij} is the usual Kronecker symbol and

$$\alpha = \tfrac{1}{4}(1 - 4m^2), \qquad \beta = \tfrac{1}{4}(l+1)(l+2), \qquad \gamma = \tfrac{1}{4}l(l-1), \qquad a_0 = l + \tfrac{1}{2} \tag{2.35}$$

Using these values, one can compute from (2.32) the values given on Table III. With the D's, E's and F's given as above, we now define

$$\psi_{m,n}^2(x) = \frac{e^{cy}(1-y)^{l/2}}{\sqrt{y}(1+y)^{(l+1)/2}}\left[1 + \sum_{j=1}^{\infty} \frac{1}{(2c)^j} f_j(y)\right] \tag{2.36}$$

with $f_j(y)$ given by (2.30) and $y = \sqrt{1-x^2}$.

For any fixed y with $0 < y < 1$, any given term in the sum (2.36) can be made small compared to 1 by choosing c sufficiently large. For fixed c, however, due to the terms in y^{-j} and $(1-y)^{-j}$ in f_j as given by (2.30), the j^{th} term in (2.36) will be small compared to unity only if $(2c)^{-1} \ll y \ll 1 - (2c)^{-1}$, or, what is the same, if $c^{-\frac{1}{2}} \ll x \ll 1 - (8c^2)^{-1}$. For fixed large c then, the first few terms of (2.36) will describe ψ accurately if $c^{-\frac{1}{3}} \leqq x \leqq 1 - c^{-1}$ which is the restriction we have imposed.

2.4 *Joining of the Solution* $\psi_{m,n}^1$ *and* $\psi_{m,n}^2$. In terms of the variable x, our solution (2.12) reads

$$\psi_{m,n}^1(x) = (1-x^2)^{m/2} \sum_{i=0}^{\infty} \sum_{j=-2i}^{2i} \frac{1}{(2c)^i} A_j^i D_{l+2j}(x\sqrt{2c}). \tag{2.37}$$

In this series, substitute

$$x = u/c^{\alpha} \tag{2.38}$$

For fixed u and large c, the argument of the Weber function in (2.37), namely $x\sqrt{2c} = u\sqrt{2}c^{\frac{1}{2}-\alpha}$ becomes large if $\alpha < \frac{1}{2}$. The ratio of the $(i+1)^{\text{st}}$ term to the i^{th} term is now $O(c^{1-4\alpha})$ under these assumptions, so that if $\alpha > \frac{1}{4}$, the successive terms of (2.12) will be ordered in powers of $1/c^{4\alpha-1}$ and the first term of the sum can be expected to give a good value for ψ for large enough c. Accordingly we inquire into the limiting form of (2.37) when (2.38) holds with fixed u, large c, and

$$\tfrac{1}{4} < \alpha < \tfrac{1}{2}. \tag{2.39}$$

We now substitute into (2.37) the asymptotic expression

$$D_\nu(u\sqrt{2}c^{\frac{1}{2}-\alpha}) \sim [u\sqrt{2}c^{\frac{1}{2}-\alpha}]^\nu e^{-\frac{1}{4}u^2 2c^{1-2\alpha}} \sum_{k=0} \frac{(-1)^k[\nu]_{2k}}{k!\,u^{2k}c^{(1-2\alpha)k}2^{2k}} \tag{2.40}$$

where the falling factorial $[k]_j$ is $k(k-1)\cdots(k-j+1)$ for positive integral j, one for $j = 0$, and zero for all other values of j. (See [16], Vol. II, Eq. (8.4.1), p 122.) The resultant triple sum can be rearranged to yield

$$\psi_{m,n}^1(u/c^\alpha) = 2^{l/2}c^{\beta l/2}u^l e^{-u^2c^\beta/2}\left[1 - \frac{u^2}{c^{2\alpha}}\right]^{m/2} \times \sum_{i=0}^{\infty}\sum_{\nu=-2i}^{\infty}\frac{1}{c^{i+\beta\nu}}\sum_{j=-2i}^{2i}\frac{(-1)^{j+\nu}[l+2j]_{2(\nu+j)}A_j^i}{u^{2\nu}2^{i+j+2\nu}(\nu+j)!} \tag{2.41}$$

where $\beta = 1 - 2\alpha$. The expression in brackets can be expanded, and one sees that, as functions of u, the terms of the expanded form for ψ^1 are all of the form $Au^p e^{-u^2 c^\beta/2}$. For large c, we see from (2.41) that the term with $p = l$ dominates, since the only term in the triple sum that does not involve c to a negative exponent (namely the term $i = \nu = 0$) has the value unity.

We now write

$$\psi^1_{m,n}(u/c^\alpha) \to 2^{l/2} c^{\beta l/2} u^l e^{-u^2 c^\beta/2} g_1(c) \tag{2.42}$$

where $g_1(c)$ is obtained from (2.41) by extracting the terms independent of u from the product of the triple sum and the expression in brackets. Straightforward manipulations finally yield $g_1(c) = 1 + g_1^1/c + g_2^1/c^2 + \cdots$ with

$$g_k^1 = \sum_{i=0}^{k} \sum_{j=-2i}^{2i} \frac{(-1)^j [m/2]_{k-i} \, [l + 2j]_{2(k-i+j)} \, A_j^i}{(k-i)!(k-i+j)! \, 2^{2k+j-i}} .$$

Note that $g_1(c)$ does not depend upon α provided (2.39) holds. The first few g_k^1 as computed by ALPAK are given on Table IV.

We next consider our solution $\psi^2_{m,n}(x)$ as given by (2.36) and (2.30) when $x = u/c^\alpha$. The ratio of the $(i + 1)^{\text{st}}$ term in (2.36) to the i^{th} term is now $O(c^{2\alpha-1})$, so that if $\alpha < \frac{1}{2}$, (as is required by (2.39)), the first term of the sum can be expected to give a good value for ψ for large enough c.

We now expand the various factors of $\psi^2_{m,n}(u/c^\alpha)$. Since $y = (1 - x^2)^{\frac{1}{2}} = (1 - u^2/c^{2\alpha})^{\frac{1}{2}}$, we can write

$$\begin{aligned} e^{cy} &= \exp\left\{c\left[1 - \frac{1}{2\cdot 1!}\frac{1}{c^{2\alpha}} - \frac{1\cdot 1}{2^2\cdot 2!}\frac{u^4}{c^{4\alpha}} - \cdots\right]\right\} \\ &= e^c \, e^{-u^2 c^\beta/2}\left[1 + \sum_{i=2}^{\infty} u^{2i} \sum_{j=1}^{[i/2]} \frac{R_{ij}(-1)^j}{c^{2\alpha i - j}}\right] \end{aligned}$$

where

$$R_{ij} = \sum \frac{a_2^{s_2} \, a_3^{s_3} \cdots a_i^{s_i}}{s_2 ! \, s_3 ! \cdots s_i !}, \qquad a_j = \frac{1\cdot 3\cdot 5 \cdots (2j-3)}{2^j j!},$$

and the sum defining R_{ij} is over all partitions $2s_2 + 3s_3 + \cdots + is_i = i$ of i into j parts with no part less than two. It is convenient to extend the definition of R_{ij} so that $R_{0j} = \delta_{j0} = R_{j0}$, $R_{1j} = 0$.

For the remaining part of the factor in (2.36), one finds

$$\frac{1}{\sqrt{y}} \frac{(1 - y)^{l/2}}{(1 + y)^{(l+1)/2}} = \frac{u^l}{c^{l\alpha} 2^{l+\frac{1}{2}}} \sum_{i=0} M_i \frac{u^{2i}}{c^{2i\alpha}}$$

with

$$M_0 = 1, \qquad M_1 = (2l + 3)/2^3, \qquad M_2 = (4l^2 + 24l + 31)/2^6\cdot 2!,$$

$$M_3 = (8l^3 + 108l^2 + 454l + 561)/2^9\cdot 3!,$$

$$M_4 = (16l^4 + 384l^3 + 3320l^3 + 12000l + 14577)/2^{12}\cdot 4!,$$

etc.

Combining these results, we find

$$\psi^2_{m,n}\left(\frac{u}{c^\alpha}\right) = \frac{e^c u^l e^{-u^2 c^\beta/2}}{c^{\alpha l} 2^{l+\frac{1}{2}}}\left[1 + \sum_{j=1} \frac{1}{(2c)^j} f_j(y)\right] \sum_{i=0}^{\infty}\sum_{k=0}^{\infty}\sum_{j=0}^{[i/2]} \frac{u^{2(i+k)} M_k R_{ij}(-1)^j}{c^{2\alpha(i+k)-j}} \tag{2.43}$$

The expression in brackets here can be expanded, and the resultant series for ψ^2 will consist of terms of the form $Bu^p e^{-u_2 c^\beta/2}$. Equation (2.43) shows that the term with $p = l$ dominates for large c, and so we write

$$\psi^2_{m,n}(u/c^2) \rightarrow \frac{e^c u^l e^{-u^2 c^\beta/2}}{c^{\alpha l} 2^{l+\frac{1}{2}}}\, g_2(c) \tag{2.44}$$

where $g_2(c)$ consists of all terms independent of u that are obtained from the product of the triple sum and expression in brackets in (2.43). These terms can be sorted out in a routine manner. We find, after much expanding and rearranging that $g_2(c) = 1 + g_1^2/c + g_2^2/c^2 + \cdots$ with

$$2^j g_j^2 = \sum_{k=1}^{j}\left[D_k^j + \frac{E_k^j}{2^k} + F_k^j\, U_{kk}\right] + \sum_{r=1}^{j} M_r \sum_{q=r}^{j} F_q^j\, U_{qq-r} + \sum_{k=1}^{j}\sum_{r=2k}^{k+j}\sum_{q=r}^{k+j}\sum_{i=2k}^{r} \frac{F_q^{k+j} U_{qq-r}\, M_{r-i}\, R_{ik}(-1)^k}{2^k}.$$

Here the U's are given by

$$(1 - y)^{-q} = \sum_{p=0} U_{qp}\, \frac{c^{2\alpha(q-p)}}{u^{2(q-p)}}$$

so that $U_{q0} = 2^q$, $U_{q1} = -q2^{q-2}$, $U_{q2} = q(q - 3)2^{q-4}/2!$, etc. Note that $g_2(c)$ is independent of α provided this quantity is less than $\frac{1}{2}$ The first few g_j^2 as computed by ALPAK are given on Table IV.

The solution ψ^1 and ψ^2 should differ only by a constant factor (possibly a function of c). Comparison of (2.42) and (2.44) shows that for large c and $x = u/c^\alpha$, u fixed and $\frac{1}{4} < \alpha < \frac{1}{2}$, the lead term of $\psi^1_{m,n}$ and $k_2\psi^2_{m,n}$ will be identical if we choose

$$k_2 = \frac{2^{l/2} c^{\beta l/2} g_1(c)}{e^c g_2(c)/c^{\alpha l} 2^{l+\frac{1}{2}}} = \frac{e^{-c} c^{l/2} 2^{(3l+1)/2} g_1(c)}{g_2(c)} \tag{2.45}$$

as reported in (1.11). Note that this is independent of α.

2.5 *Solution for* $1 - c^{-1} \leqq x \leqq 1$. We seek a solution of (2.18) near $y = 0$ valid for large c. Substitute $y = s/c$ into (2.18) to obtain

$$(c^2 - s^2)\frac{d^2\psi}{ds^2} + \left(\frac{c^2}{s} - 2s\right)\frac{d\psi}{ds} + \left(\chi - c^2 + s^2 - \frac{m^2 c^2}{s^2}\right)\psi = 0.$$

Into this equation substitute the series (2.13) for χ. We write $a_0 = l + \frac{1}{2}$. Regrouping the terms results in

$$\mathfrak{M}\psi = \frac{1}{(2c)^2}\,\mathfrak{N}\psi = \sum_{j=0}^{\infty} \frac{a_j}{(2c)^{j+1}}\,\psi \tag{2.46}$$

with the operators given by

$$4\mathfrak{M} = \frac{d^2}{ds^2} + \frac{1}{s}\frac{d}{ds} - \left(1 + \frac{m^2}{s^2}\right) \tag{2.47}$$

$$\mathfrak{N} = s^2 \frac{d^2}{ds^2} + 2s \frac{d}{ds} - s^2 \tag{2.48}$$

Equation (2.46) suggests solution in the form

$$\psi = \sum_{j=0}^{\infty} \frac{1}{(2c)^j} h_j(s) \tag{2.49}$$

with the h's independent of c. By substituting this expression in (2.46) and equating to zero the coefficients of distinct powers of $1/c$, we find the system of differential equations

$$\begin{aligned} \mathfrak{M}h_0 &= 0, \qquad \mathfrak{M}h_1 = -a_0 h_0 \\ \mathfrak{M}h_j &= \mathfrak{N}h_{j-2} - \textstyle\sum_{i=0}^{j-1} a_i h_{j-1-i}, \quad j = 2, 3, \cdots. \end{aligned} \tag{2.50}$$

We are, of course, seeking a solution of (2.46) that remains bounded at $s = 0$ for all c, and hence require each h to be bounded at $s = 0$ We are forced then to take from the first of (2.50) $h_0(s) = I_m(s)$ where I_m is the usual Bessel function of pure imaginary argument ([16], Vol. II, p. 5).

To display explicitly a solution for the remaining equations of (2.50), it is convenient to introduce the functions

$$P_j(s) = s^j I_{m+j}(s) \qquad j = 0, 1, 2, \cdots. \tag{2.51}$$

From (2.47), (2.48) and the recurrence relations and differential equation for the I's ([16], Vol. II, p. 79) one readily establishes the formulae

$$\begin{aligned} \mathfrak{M}P_j &= \tfrac{1}{2} j P_{j-1} \\ \mathfrak{N}P_j &= [2j(2j+1) + m(4j+1) + m^2]P_j + (2j+1)P_{j+1}, \\ &\qquad\qquad j = 0, 1, 2, \cdots. \end{aligned} \tag{2.52}$$

Examination of (2.50) now shows that constants B_k^j can be chosen so that

$$h_0(s) = P_0(s) = I_m(s), \qquad h_j(s) = \textstyle\sum_{i=1}^{j} B_i^j P_i(s), \quad j = 1, 2, \cdots. \tag{2.53}$$

Indeed, substituting this assumed solution into (2.50), we find

$$\begin{aligned} \tfrac{1}{2}B_1^1 &= -a_0 = -(l + \tfrac{1}{2}), \qquad \tfrac{1}{2}B_1^2 = m^2 + m - a_1, \\ &\tfrac{1}{2}\cdot 2B_2^2 = 1 - a_0 B_1^1 \end{aligned} \tag{2.54}$$

and for $j = 3, 4, \cdots$ and $i = 1, 2, 3, \cdots, j$

$$\begin{aligned} \tfrac{1}{2} i B_i^j = [2(i-1)(2i-1) + m(4i-3) + m^2]B_{i-1}^{j-2} + (2i-3)B_{i-2}^{j-2} \\ - \textstyle\sum_{k=0}^{j-i} a_k B_{i-1}^{j-1-k} \end{aligned} \tag{2.55}$$

Here we define $B_i^0 = \delta_{i0}$ and for $j \neq 0$ we define $B_i^j = 0$ if $i > j$ or $i < 1$ The h's are thus completely determined.

We now write

$$\psi_{m,n}^3(x) = I_m(s) + \sum_{j=1}^{\infty} \frac{1}{(2c)^j} \sum_{k=1}^{j} B_k^j \, s^k I_{m+k}(s) \tag{2.56}$$

$$s = cy, \qquad y = \sqrt{1 - x^2}, \qquad |x| \leqq 1$$

with the B's given by (2.54)–(2.55) and the a's by (2.13). The first few B's are given in Table II. Note that when $x = 1$, i.e., $y = s = 0$, $\psi_{m,n}^3$ is zero if $m \neq 0$, whereas $\psi_{0,n}^3(x = 1) = 1$.

2.6 *Joining of the Solutions* $\psi_{m,n}^2$ *and* $\psi_{m,n}^3$. As expressed in terms of the variable s, the right side of (2.56) appears ordered in inverse powers of c. If, however, $s = cy$ is substituted there, the terms are no longer ordered in powers of $1/c$, so that the first few terms of (2.56) give useful information about ψ only if y tends to zero as c gets large.

We observe now that if, with v fixed, $y = v/c^\alpha$, then $s \to \infty$ and the terms of (2.56) are ordered in powers of $1/c^\alpha$ provided $0 < \alpha < 1$. The argument of the Bessel function in (2.56) is now large and we can use the well-known asymptotic form ([16], Vol. II, Eq (7.13.5), p 86)

$$I_\nu(z) = \frac{e^z}{\sqrt{2\pi z}} \sum_{j=0} \frac{(-1)^j(\nu, j)}{(2z)^j} \tag{2.57}$$

where $(\nu, 0) = 1$ and for $j \neq 0$,

$$(\nu, j) = \frac{(4\nu^2 - 1)\,(4\nu^2 - 3^2) \cdot \cdot [4\nu^2 - (2j - 1)^2]}{2^{2j} j!} .$$

Inserting this expression in (2.56) and rearranging the sum, we find

$$\psi_{m,n}^3(y = v/c^\alpha) = \frac{e^{vc^{1-\alpha}}}{c^{(1-\alpha)/2}\sqrt{2\pi v}} \sum_{j=0}^{\infty} \sum_{k=0}^{j} \sum_{i=0}^{\infty} \frac{(-1)^i(m + k, i) B_k^j \, v^{k-i}}{2^{j+i} c^{j+(i-k)(1-\alpha)}} . \tag{2.58}$$

As functions of v, the terms of the series (2.58) are all of the form $Av^{p-\frac{1}{2}} e^{vc^{1-\alpha}}$. For large c, the term with $p = 0$ dominates. We write

$$\psi_{m,n}^3(y = v/c^\alpha) \to \frac{e^{vc^{1-\alpha}}}{c^{(1-\alpha)/2}\sqrt{2\pi v}} h_3(c) \tag{2.59}$$

where $h_3(c) = 1 + h_1^3/c + h_2^3/c^2 + \cdot \cdot$ consists of those terms of the triple sum in (2.58) that are independent of v. We find

$$h_j^3 = \sum_{i=0}^{j} \frac{B_i^j(-1)^i(m + i, i)}{2^{i+j}} \tag{2.60}$$

so that $h_3(c)$ is independent of α. Some values of h_j^3 are given in Table IV.

We now investigate the behavior for large c of $\psi_{m,n}^2(y = v/c^\alpha)$ as given by (2.36). Since $\alpha > 0$, $y \to 0$ for large c and we have

$$\psi^2_{m,n}\left(y = \frac{v}{c^\alpha}\right) = \frac{e^{vc^{1-\alpha}} c^{\alpha/2}}{\sqrt{v}}\left[1 + \sum_{j=1}^\infty \frac{1}{(2c)^j} f_j(y)\right] \times \sum_{k=0}^\infty \frac{v^k}{c^{\alpha k}} \sum_{i=0}^k (-1)^i \left[\frac{l}{2}\right]_i \left[-\frac{l+1}{2}\right]_{k-i} \tag{2.61}$$

where as before the subscripted bracket is a falling factorial. As functions of v, the terms of this series are all of the form $Av^{p-\frac{1}{2}}e^{vc^{1-\alpha}}$. We see from (2.61) that for large c, the term with $p = 0$ dominates. We write

$$\psi^2_{m,n}(y = v/c^\alpha) \to \frac{c^{\alpha/2} e^{vc^{1-\alpha}}}{\sqrt{v}} h_2(c) \tag{2.62}$$

where $h_2(c) = 1 + h_1^2/c + h_2^2/c^2 + \cdots$ consists of those terms independent of v obtained from the product of the triple sum of (2.61) with the expansion of the square bracket factor there. Straightforward manipulations yield

$$2^j h_j^2 = \sum_{k=1}^j \left(E_k^j + F_k^j + D_k^j \sum_{\sigma=0}^k \frac{(-1)^\sigma [l/2]_\sigma\, [-\frac{1}{2}(l+1)]_{k-\sigma}}{\sigma!(k-\sigma)!}\right). \tag{2.63}$$

Note that $h_2(c)$ is independent of α. Some values of h_j^2 are given in table IV.

Comparison of (2.59) and (2.62) now show that for large c and $y = v/c^\alpha$, $0 < \alpha < 1$, the lead terms of $k_2\psi^2$ and $k_3\psi^3$ will be identical if

$$k_3 = k_2 \frac{c^{\alpha/2} c^{(1-\alpha)/2} \sqrt{2\pi v}\, h_2(c)}{\sqrt{v}\, h_3(c)} = \sqrt{2\pi c}\, \frac{h_2(c)}{h_3(c)}\, k_2$$

as reported in (1.12) and (1.15).

2.7 *Solution for* $1 \leqq x \leqq 1 + c^{-1}$. When $|x| \geqq 1$, we adopt the abbreviation $z = \sqrt{x^2 - 1}$. The substitution $y = iz$ in (2.56) then yields

$$\psi_{m,n}(x) = i^m \left[J_m(cz) + \sum_{j=1}^\infty \frac{1}{(2c)^j} \sum_{k=1}^j B_k^j\, (-cz)^k J_{m+k}(cz)\right]$$

as a formal solution of (1.1) valid for $1 \leqq x \leqq 1 + c^{-1}$. Here J_ν is the usual Bessel function and we have used the relation $I_\nu(iz) = i^\nu J_\nu(z)$. We must now choose an appropriate constant multiplier for this expression to join it to ψ^3_{mn}.

It is well known ([1], p. 230) that the eigenfunctions of (1.1) can be written in the form $\psi(x) = (1 - x^2)^{m/2}\phi(x)$ where $\phi(x)$ is the restriction to the real line of an entire function. If then m is even, ψ itself is entire and it is natural to join a solution valid for $|x| < 1$ to its analytic continuation for $x > 1$. If m is an odd integer, and $\psi = (1 - x^2)^{m/2}\phi(x)$ is a solution valid for $|x| < 1$, we follow Flammer ([2], bottom page page 32) in taking $\psi = (x^2 - 1)^{m/2}\phi(x)$ as the extension of this solution valid for $x > 1$. Here ϕ is to be continued analytically.

With these considerations we write

$$\begin{aligned}\psi^4_{m,n}(x) &= \frac{\epsilon_m}{i^m} \psi^3_{m,n}(x)|_{y=iz} \\ &= \epsilon_m \left[J_m(cz) + \sum_{j=1}^\infty \frac{1}{(2c)^j} \sum_{k=1}^j B_k^j(-cz)^k J_{m+k}(cz)\right],\end{aligned} \tag{2.64}$$

with $z = \sqrt{x^2 - 1}$ and $\epsilon_m = 1$ if m is odd and $(-1)^{m/2}$ if m is even.

2.8 *Solution for* $1 + c^{-1} \leqq x$ Examination of the formalism of Section 2.3 shows that the substitution $y = iz = i\sqrt{x^2 - 1}$ in ψ^2 yields a complex formal solution of (1.1) valid for $x > 1$. We multiply this solution by a complex constant and take its real part as ψ^5 Explicitly,

$$\begin{aligned} \psi^5_{m,n}(x) &= \operatorname{Re} e^{-im\pi/2}\, \psi^2_{m,n}(x)|_{y=iz} \\ &= \operatorname{Re} \frac{e^{i(cz - m\pi/2)}(1 - iz)^{l/2}}{\sqrt{iz}(1 + iz)^{(l+1)/2}} \left[1 + \sum_{j=1} \frac{1}{(2c)^j} f_j(iz)\right] \end{aligned} \tag{2.65}$$

where the f_j are given by (2.30) and (2.32).

The joining factor for ψ^4 and ψ^5 is obtained by setting $z = w/\sqrt{c}$ in (2.64) and (2.65). The expansions are similar to those used to join ψ^2 and ψ^3. One finds $\sqrt{\pi c}h_2(c)\psi^4 = \sqrt{2}\, h_3(c)\psi^5$ as reported in (1.13).

For very large x, the solution (2.65) becomes

$$\psi^5_{m,n}(x) \sim \frac{\cos\,[cx - (n + 1)\pi/2]}{x}.$$

Flammer's radial function [2], p. 31, is related by $cR^{(1)}_{m,n}(c, x) = \psi^5_{m,n}(x)$ whence (1.21).

2.9 *The Normalization Factor.* We now seek a normalization factor $N_{m,n}$ for the solution (1.4). That is we require

$$\int_{-1}^{1} [N_{m,n}\, \hat{\psi}_{m,n}(x)]^2\, dx = 1.$$

We of course use the fact that the eigenfunctions of (1.1) are even or odd according as l is even or odd to extend (1.4) to $|x| \leqq 1$.

The contribution from ψ^1 to $\int \hat{\psi}^2 dx$ is

$$\begin{aligned} \int_0^{c^{-1/3}} dx\, [\psi^1_{m,n}(x)]^2 &= \frac{1}{\sqrt{2c}}\int_0^\infty dt\, [\psi^1_{m,n}(t/\sqrt{2c})]^2 - \frac{1}{\sqrt{2c}}\int_{c^{1/6}2^{1/2}}^\infty dt [\psi^1_{m,n}(t/\sqrt{2c})]^2 \\ &\equiv J_1 - J_2 . \end{aligned}$$

From

$$\psi^1_{m,n}(t/\sqrt{2c}) = \left(1 - \frac{t^2}{2c}\right)^{m/2} \sum_{j=0} \frac{1}{(2c)^j} \sum_{k=2j}^{2j} A^j_k\, D_{l+2k}(t)$$

it is clear that $[\psi^1(t/\sqrt{2c})]^2$ can be developed in powers of $1/c$. Terms such as $t^{2j}D_\rho(t)D_\sigma(t)$ can be expressed as sums of products of D's by repeated use of the recurrence

$$tD_\nu(t) = \nu D_{\nu-1}(t) + D_{\nu+1}(t).$$

To carry out the integration implied in J_1, one then needs only the formula

$$\int_0^\infty [D_\nu(t)]^2\, dt = \nu!\, \sqrt{\pi/2}$$

We belabor the point no further. The first few terms of $2J_1$ appear on the right of (1.16). In a similar development for J_2, one finds each term to be $O(c^p e^{-\sqrt{c}}$ for some finite p.

Simple arguments show that the contribution to N^2 due to integration of the square of (1.4) over the range $c^{-\frac{1}{4}} \leqq x \leqq 1$ is $O(e^{-c^{\alpha}})$ for $0 < \alpha < \frac{1}{2}$. We have then finally $N_{n,m}^2 = \frac{1}{2}J_1$ as reported in (1.16).

The normalization factor $C_{m,n}$ connecting Flammer's functions with our ψ_{mn}^1, $\psi_{mn}^1(x) = C_{m,n}S_{m,n}(c, x)$, can be obtained by evaluating this equation or its derivative at $x = 0$ using (2.37) and the formulae

$$D_{2p}(0) = \frac{(-1)^p(2p)!}{p!\,2^p}, \qquad \frac{d}{dt}D_{2p+1}(t)|_{t=0} = \frac{(-1)^p(2p+1)!}{2^p q!}.$$

The results are stated in (1.19)–(1.20).

III. Asymptotics for n and c Both Large 3.1 *The Oscillatory Regions of* $\psi_{m,n}(x)$. When n and c both are large, with $n = (2/\pi)c$ say, numerical evidence indicates that $\chi_{n,m} \sim c^2$ Let us write

$$\chi_{n,m} = c^2 + 2\delta c + \sum_{j=0}^{\infty} \frac{b_j}{c_j}. \tag{3.1}$$

When this expression is used, the roots of (2.16) become for large c and $m \neq 1$, $x = 1 + [\delta \pm \sqrt{\delta^2 + m^2 - 1}]/c$. If $m > 1$, the two roots surround $x = 1$ and converge to it as c increases The eigenfunction is oscillatory except in this small region surrounding $x = 1$. If $m = 0$ and $|\delta| < 1$, the roots are complex and $r(x)$ of (2.14) is always positive. If $|\delta| > 1$, again r is negative only in a small region surrounding $x = 1$. Finally, if $m = 1$, we see from (2.15) that $r = (\chi - c^2x^2)/(1 - x^2)$ and r is negative only in an interval of width δ/c, one endpoint of which is $x = 1$.

These considerations lead us to expect that as long as (3.1) holds, ψ will be oscillatory everywhere. Its behavior around $x = 1$, must of course be examined separately.

3.2 *Solution near* $x = 1$ In (1.1) make the substitutions

$$\psi = (1 - x^2)^{m/2}U(x) \tag{3.2}$$

and $\eta = 1 - x$ to obtain

$$\eta(2 - \eta)\frac{d^2U}{d\eta^2} + 2(m + 1)(1 - \eta)\frac{dU}{d\eta} + [\chi - m^2 - m - c^2(1 - 2\eta + \eta^2)]U = 0. \tag{3.3}$$

Examination of the indicial equation associated with (3.3) shows that at $x = 1$, i.e., $\eta = 0$, one of the two independent solutions is bounded. The other has a logarithmic singularity if $m = 0$ and if $m \neq 0$ behaves as $(1 - x)^{-m}$. From

(3.2) then we see that in order to obtain a ψ bounded at $x = 1$ we must take for U the unique (apart from a constant factor) solution of (3.3) bounded at $\eta = 0$. With c fixed, this solution depends on the single parameter χ. We now seek an asymptotic form for this solution

Set $\eta = i\xi/c$,

$$U = e^{ic\eta}V \tag{3.4}$$

and replace χ by the expansion (3.1). Equation (3.3) then gives for V

$$-4ic\mathfrak{Q}V = \mathfrak{R}V - V\sum_{j=0}\frac{b_j}{c^j} \tag{3.5}$$

where

$$\mathfrak{Q}V = \xi\frac{d^2V}{d\xi^2} + (m+1-\xi)\frac{dV}{d\xi} - \frac{1}{2}(m+1-i\delta)V \tag{3.6}$$

and

$$\mathfrak{R}V = \xi^2\frac{d^2V}{d\xi^2} + \xi\,[2(m+1)-\xi]\frac{dV}{d\xi} + [m^2+m-(m+1)\xi]\,V. \tag{3.7}$$

Write

$$V = V_0 + \sum_{j=1}\frac{1}{c^j}V_j\,. \tag{3.8}$$

Substitute this expression in (3.5) and equate to zero the coefficients of distinct powers of c. There results the system of differential equations

$$\mathfrak{Q}V_0 = 0, \qquad -4i\mathfrak{Q}V_j = \mathfrak{R}V_{j-1} - \textstyle\sum_{k=0}^{j-1} b_k V_{j-k-1}\,, \quad j = 1, 2, \tag{3.9}$$

for the determination of V.

Now the solution of $\mathfrak{Q}V_0 = 0$ that is bounded for $\xi = 0$ is

$$V_0(\xi) = \Phi(\beta, m+1, \xi) \tag{3.10}$$

where for brevity we have set

$$\beta = \tfrac{1}{2}(m+1-i\delta) \tag{3.11}$$

and

$$\Phi(a, c, x) = 1 + \frac{a}{c}\frac{x}{1!} + \frac{a(a+1)}{c(c+1)}\frac{x^2}{2!} + \frac{a(a+1)(a+2)}{c(c+1)(c+2)}\frac{x^3}{3!} + \cdots$$

is the confluent hypergeometric function in the notation of [16], Vol. 1, Chapter 6. The lead term in our solution is now

$$\psi(x) = (1-x^2)^{m/2}e^{ic(1-x)}\Phi(\beta, m+1; -2ic(1-x)). \tag{3.12}$$

We now construct a solution for the remaining equations of (3.9). Let us define $T_p(\xi) = \Phi(p + \beta, 2p + m + 1, \xi)$ and note that $T_0(\xi) = V_0(\xi)$ by (3.10). Then from the known recurrence relations for the confluent hypergeometric function ([16], Vol. 1, p. 254), one can derive the equations

$$\xi \frac{d}{d\xi} T_p = \left[-(2p + m) + \frac{p + m - \beta}{2p + m - 1} \xi \right] T_p + (2p + m) T_{p-1} \tag{3.13}$$

$$\frac{d}{d\xi} T_p = \frac{p + \beta}{2p + m + 1} T_p + \frac{(p + \beta)(p + m + 1 - \beta)}{(2p + m + 1)^2 (2p + m + 2)} \xi T_{p+1} \tag{3.14}$$

$$\begin{aligned}(2p + m) T_{p-1} + (2p + m) \left[-1 + \frac{1 + m - 2\beta}{(2p + m)^2 - 1} \xi \right] T_p \\ - \frac{(p + \beta)(p + m + 1 - \beta)}{(2p + m + 1)^2 (2p + m + 2)} \xi^2 T_{p+1} = 0\end{aligned} \tag{3.15}$$

and of course one has $\xi T''_p + (2p + m + 1 - \xi) T'_p - (p + \beta) T_p = 0$. From these equations and the definitions (3.6) and (3 7) one can show that

$$\mathcal{Q}\xi^q T_p = [q(q + m)\xi^{q-1} + (p - q)\xi^q] T_p + 2(q - p)\xi^q T'_p \tag{3.16}$$

$$\begin{aligned}\mathcal{R}\xi^q T_p = {} & [(q + m)(q + m + 1)\xi^q + (p - q + \beta - m - 1)\xi^{q+1}] T_p \\ & + [2(q - p) + m + 1]\xi^{q+1} T'_p.\end{aligned} \tag{3.17}$$

Specializing the foregoing we find

$$\begin{aligned}
\mathcal{Q}\xi^q T_0 &= \left[q(q + m)\xi^{q-1} + \frac{q(2\beta - m - 1)}{m + 1} \xi^q \right] T_0 \\
&\qquad + \frac{2q\beta(m + 1 - \beta)}{(m + 1)^2 (m + 2)} \xi^{q+1} T_1 \\
\mathcal{Q}\xi^{q+1} T_1 &= 2q(m + 2)\xi^q T_0 + \left[(q - 1)(q - 1 - m)\xi^q - \frac{q(2\beta - m - 1)}{m + 1} \xi^{q+1} \right] T_1 \\
\mathcal{R}\xi^q T_0 &= \left[(q + m)(q + m + 1)\xi^q + \frac{(q + m + 1)(2\beta - m - 1)}{m + 1} \xi^{q+1} \right] T_0 \\
&\qquad + \frac{(2q + m + 1)\beta(m + 1 - \beta)}{(m + 1)^2 (m + 2)} \xi^{q+2} T_1 \\
\mathcal{R}\xi^{q+1} T_1 &= \\
&\cdot (m + 2)(2q + m + 1)\xi^{q+1} T_0 + \left[q(q - 1)\xi^{q+1} - \frac{q(2\beta - m - 1)}{m + 1} \xi^{q+2} \right] T_1
\end{aligned} \tag{3.18}$$

where the first of these equations comes from (3.16) and (3.14), the second from (3.16) and (3.13), the third from (3.17) and (3.14) and the last from (3.17) and (3.13).

Returning to (3.9), we have for $j = 1$ on recalling that $V_0 = T_0$,

$$-4i\mathcal{Q}V_1 = \mathcal{R}T_0 - b_0\,T_0$$
$$= [m(m+1) - b_0 + (2\beta - m - 1)\xi]T_0 + \frac{\beta(m+1-\beta)}{(m+1)(m+2)}\,\xi^2 T_1 \qquad (3.19)$$

where we have used the third equation of (3.18) with $q = 0$ to evaluate the right member. But now from the first two equations of (3.18) it is seen that $\mathcal{Q}$ operating on a linear combination of ξT_0 and $\xi^2 T_1$ will have terms of the same form as the right of (3.19). Indeed,

$$-4i\mathcal{Q}(K_1^1\,\xi T_0 + L_1^1\,\xi^2 T_1) = -4i\left[\left\{(m+1)K_1^1 + \left(K_1^1\,\frac{2\beta - m - 1}{m+1}\right.\right.\right.$$
$$\left.\left.\left. + L_1^1\,2(m+2)\right)\xi\right\}T_0 + \left\{K_1^1\,\frac{2\beta(m+1-\beta)}{(m+1)^2(m+2)} - L_1^1\,\frac{2\beta - m - 1}{m+1}\right\}\xi^2 T_1\right].$$

Comparison with (3.19) shows that

$$V_1(\xi) = \frac{i}{4}\,\frac{[\beta^2 + (m+1-\beta)^2]}{m+1}\,\xi T_0 + \frac{i}{4}\,\frac{2\beta(2\beta - m - 1)(m+1-\beta)}{(m+1)^2\,(m+2)}\,\xi^2 T_1 \qquad (3.20)$$

is a solution of (3.19) provided we choose

$$b_0 = m(m+1) - \beta^2 - (m+1-\beta)^2 = \tfrac{1}{2}(\delta^2 + m^2 - 1). \qquad (3.21)$$

A little study of (3 9) and (3.18) shows that this scheme can be continued. If we set

$$V_j(\xi) = \textstyle\sum_{k=1}^{j} K_k^j \xi^k T_0(\xi) + \sum_{k=1}^{j} L_k^j \xi^{k+1} T_1(\xi) \quad j = 1, 2, \cdots \qquad (3.22)$$

the K's, L's and b's can be successively determined to satisfy (3.9). Each b will be a polynomial in δ. Combined with (3.8) and (3.4) we will then have found a formal solution of (3.3) bounded at $\xi = \eta = 0$. This solution depends on the single parameter δ which is related to χ by (3.1). We have then found the one parameter family of bounded solutions of (3.3) referred to just below that equation. The point here is that our method of solution of the system (3.9) appears somewhat arbitrary especially with regard to the determination of the b's. Yet one can only expect a one parameter family of bounded solutions of (3.3) and we have found it.

To show how the K's, L's and b's can be determined by a recurrence scheme, we first write (3.18) with obvious abbreviations as

$$\mathcal{Q}\xi^{q+p}T_p = [a_p(q)\xi^{q-1} + b_p(q)\xi^q]T_0 + [c_p(q)\xi^q + d_p(q)\xi^{q+1}]T_1$$
$$\mathcal{R}\xi^{q+p}T_p = [e_p(q)\xi^q + f_p(q)\xi^{q+1}]T_0 + [g_p(q)\xi^{q+1} + h_p(q)\xi^{q+2}]T_1, \qquad (3.23)$$
$$p = 0, 1$$

Now substitute (3.22) into the general equation of (3 9) and use (3 23) The resulting expression is an identity in ξ provided

$$\begin{aligned}
&- 4i[b_0(q)K_q^j + a_0(q+1)K_{q+1}^j + b_1(q)L_q^j] \\
&\quad = f_0(q-1)K_{q-1}^{j-1} + e_0(q)K_q^{j-1} + f_1(q-1)L_{q-1}^{j-1} \\
&\quad\quad - \textstyle\sum_{k=0}^{j-q-1} b_k K_q^{j-k-1} - 4i[d_0(q)K_q^j + d_1(q)L_q^j + c_1(q+1)L_{q+1}^j] \\
&\quad = h_0(q-1)K_{q-1}^{j-1} + h_1(q-1)L_{q-1}^{j-1} \\
&\quad\quad\quad + g_1(q)L_q^{j-1} - \textstyle\sum_{k=0}^{j-q-1} b_k L_q^{j-k-1}, \qquad q = 1, 2, \cdot\ , j \\
&\quad -4ia_0(1)K_1^j = \delta_{1j}e(0) - b_{j-1}, \qquad j = 1, 2, \quad .
\end{aligned} \tag{3 24}$$

Here we have adopted the conventions $L_j^0 = 0$, $K_j^0 = \delta_{j0}$, $j = \cdots -1, 0, 1,$ and for $k \neq 0$, $L_j^k = K_j^k = 0$ if $j < 1$ or $j > k$. We also require terms preceded by $\sum_a^b$ to vanish if $a > b$.

The first pair of equations (3.24) furnishes $2j$ simultaneous equations for the $2j$ quantities K_q^j, L_q^j, $q = 1, 2, \ \cdots, j$. The right side involves K's and L's of lower superscript and the quantities $b_0, b_1, \ \cdot\ , b_{j-2}$. (For $j = 1$, no b's occur.) These can be solved and the last of (3.24) used to determine b_{j-1}, whence one can proceed to the next value of j.

The $2j$ simultaneous equations can be solved explicitly by the following technique. Introduce the two component vectors

$$\mathbf{Z}_q^j = \begin{pmatrix} K_q^j \\ L_q^j \end{pmatrix} \tag{3.25}$$

and

$$\mathbf{U}_q^j = \begin{pmatrix} f_0(q-1)K_{q-1}^{j-1} + e_0(q)K_q^{j-1} + f_1(q-1)L_{q-1}^{j-1} - \sum_{k=0}^{j-q-1} b_k K_q^{j-k-1} \\ h_0(q-1)K_{q-1}^{j-1} + h_1(q-1)L_{q-1}^{j-1} + g_1(q)L_q^{j-1} - \sum_{k=0}^{j-q-1} b_k L_q^{j-k-1} \end{pmatrix} \tag{3.26}$$

and define the matrices

$$A(q) = -4i \begin{pmatrix} b_0(q) & b_1(q) \\ d_0(q) & d_1(q) \end{pmatrix}, \qquad C(q) = -4i \begin{pmatrix} a_0(q) & 0 \\ 0 & c_1(q) \end{pmatrix}. \tag{3.27}$$

The system (3.24) becomes

$$\begin{aligned}
A(j)\mathbf{Z}_j^j &= \mathbf{U}_j^j \\
A(q)\mathbf{Z}_q^j + C(q+1)\mathbf{Z}_{q+1}^j &= \mathbf{U}_q^j, \qquad q = 1, 2, \cdot\ , j-1
\end{aligned}$$

whose solution we write in the recursive form

$$\begin{aligned}
\mathbf{Z}_j^j &= A(j)^{-1}\mathbf{U}_j^j \\
\mathbf{Z}_q^j &= A(q)^{-1}[\mathbf{U}_q^j - C(q+1)\mathbf{Z}_{q+1}^j], \quad q = j-1, j-2, \cdot\ , 1 \\
b_{j-1} &= \delta_{1j}e(0) + 4ia_0(1)K_1^j, \qquad j = 1, 2, \cdots .
\end{aligned} \tag{3.28}$$

The inversion indicated here can indeed be carried out, for one finds det $A = 4iq$ and

$$A^{-1}(q) = \frac{i}{4q}\begin{pmatrix} \dfrac{-i\delta}{m+1} & 2(m+2) \\ \dfrac{(m+1)^2+\delta^2}{2(m+1)^2(m+2)} & \dfrac{i\delta}{m+1} \end{pmatrix}.$$

Table V lists values of the K's, L's and b's obtained from this recurrence by ALPAK.

The results of this section can be assembled to give $\psi^7_{m,n}(x)$ and $\psi^8_{m,n}(x)$ as presented in equations (1.25) to (1.28). The K's and L's are given by the recurrence (3.25)–(3.28) with the initial condition given below (3.24). It is easy to verify using Kummer's transformation ([16], Vol. I, p. 253) that ψ^7 and ψ^8 are real.

3.3 *Solution for* $0 \leqq x \leqq 1 - c^{-\frac{1}{3}}$. Into (1.1) substitute the expression

$$\psi_{m,n}(x) = \frac{\exp\left\{i\left[cx + \dfrac{\delta}{2}\log\left(\dfrac{1+x}{1-x}\right)\right]\right\}}{\sqrt{1-x^2}}\, w(x) \tag{3.29}$$

and the expansion (3.1). There results

$$(1-x^2)w'' + 2i[c(1-x^2) + \delta]w' + \left[\frac{1-\delta^2-m^2+2i\delta x}{1-x^2} + \sum_{j=0}\frac{b_j}{c^j}\right]w = 0,$$

or

$$-2iw' = \frac{1}{c}\,\mathcal{P}w + \frac{1}{1-x^2}\sum_{j=0}\frac{b_j}{c^{j+1}} \tag{3.30}$$

where

$$\mathcal{P}w = \frac{d^2w}{dx^2} + \frac{2i\delta}{1-x^2}\frac{dw}{dx} + \left[\frac{1-\delta^2-m^2+2i\delta x}{(1-x^2)^2}\right]w. \tag{3.31}$$

Now into (3.30) substitute the expansion

$$w = 1 + \sum_{j=1}\frac{1}{c^j}\,\phi_j(x) \tag{3.32}$$

and equate coefficients of like powers of $1/c$ to zero. The system of differential equations

$$\begin{aligned} \phi_0 &= 1 \\ -2i\phi_j' &= \mathcal{P}\phi_{j-1} + \frac{1}{1-x^2}\sum_{k=0}^{j-1} b_k\,\phi_{j-k-1}, \qquad j = 1, 2, \cdots \end{aligned} \tag{3.33}$$

results.

This system is very similar to that treated in Section 2.3. The equations can be integrated one at a time. For reasons similar to those given below (2.24) the additive constant of integration can be neglected. We find for ϕ_1, for example,

$$
\begin{aligned}
-2i\phi_1(x) &= \frac{1}{4}(1 - \delta^2 - m^2 + 2b_0)\log\frac{1+x}{1-x} \\
&\quad + \frac{1-\delta^2-m^2}{4}\left(\frac{1}{1-x} - \frac{1}{1+x}\right) + i\frac{\delta}{2}\left(\frac{1}{1-x} + \frac{1}{1+x}\right).
\end{aligned} \tag{3.34}
$$

Now the constants b_0, b_1, ..., have already been determined in the previous section Indeed, Eq. (3 21) shows that the coefficient of the logarithmic term in (3 34) vanishes, so that we can write

$$
\phi_1(x) = -\frac{\delta}{4}\left(\frac{1}{1+x} + \frac{1}{1-x}\right) + \frac{i}{4}b_0\left(\frac{1}{1+x} - \frac{1}{1-x}\right). \tag{3.35}
$$

This delightful happenstance leads one to suspect that perhaps we can write

$$
\phi_j(x) = \sum_{k=1}^{j}\left(\frac{M_k^j}{(1+x)^k} + \frac{N_k^j}{(1-x)^k}\right) \tag{3.36}
$$

for $j = 1, 2, \ldots$, i.e., that the b's as determined in the previous section will continually cause the logarithmic terms to cancel as (3.33) is integrated. The situation is now completely parallel to that treated in Section 2.3. Indeed Eqs (2 32)–(2.35) can be taken over to give a recurrence for the M's and N's by making the following substitutions and changes $u_{kj}^{1i} = d_{kj}^{1i} = D_k^j = \alpha = 0$, $i = 1, 2, 3$, $a_k = b_{k-1}$, $a_0 = -i\delta$, $\beta = -\frac{1}{2}(b_0 + i\delta)$, $\gamma = -\frac{1}{2}(b_0 - i\delta)$, $E_k^j = M_k^j$, $F_k^j = N_k^j$ The left members of (2 32) must also be multiplied by $2i$. Starting values for the recurrence are given by (3 35) Table VI gives some values of the M's and N's determined this way. In every case computed, the b's given by the analogue of (2 33) coincided with those obtained from (3.28). We have not succeeded in showing directly that these two schemes must yield the same b_k for all k

We multiply the solution (3.29), (3.32), (3.36) by the phase factor $\exp(-l\pi/2)$ and take the real part to obtain the solution $\psi_{m,n}^6(x)$ as reported in (1.24). This phase factor assures proper parity for ψ The M's and N's are given by the recurrence described above. As yet δ is unspecified. Its relation to n, m and c will be given later

3 4 *Joining of the solution* $\psi_{m,n}^6$ *and* $\psi_{m,n}^7$. In (1 24) let us substitute $x = 1 - uc^{-\frac{1}{2}}$ and for fixed u allow c to become large. By expanding in powers of $1/\sqrt{c}$ we find

$$
\begin{aligned}
\psi_{m,n}^6(x = 1 - uc^{-\frac{1}{2}}) &\sim \frac{c^{\frac{1}{4}}}{\sqrt{2u}} \\
&\cdot \operatorname{Re}\exp\left\{i\left[c - u\sqrt{c} - \frac{\delta}{2}\log\frac{u}{2\sqrt{c}} - l\frac{\pi}{2}\right]\right\}\left[1 + \sum\frac{W_j^1(u)}{(\sqrt{c})^j}\right]
\end{aligned} \tag{3 37}
$$

where

$$
\begin{aligned}
W_1^1(u) &= \frac{N_1^1}{u} + u\,\frac{1 - i\delta}{4} \\
W_2^1(u) &= \frac{N_2^2}{u^2} + \frac{1}{4}[2M_1^1 + (1 - i\delta)N_1^1] + u^2\,\frac{3 - 4i\delta - \delta^2}{32}
\end{aligned} \tag{3.38}
$$

etc.

The solution $\psi^7_{m,n}$ as given by (1.25)–(1.28) can also be developed in powers of $1/\sqrt{c}$ when $x = 1 - uc^{-1}$. Now $\xi = -2i\sqrt{c}u$ is large and we make use of the known asymptotic formula for the confluent hypergeometric functions ([16], Vol. I, p. 278). One finds the formidable expression

$$\psi^7_{m,n}(x = 1 - uc^{-1}) \sim \frac{\sqrt{2}\Gamma(m+1)e^{-\delta(\pi/4)}}{\sqrt{u}c^{(2m+1)/4}R_m(\delta)} \tag{3.39}$$
$$\operatorname{Re}\exp\left\{-i\left[u\sqrt{c} - (m+1)\frac{\pi}{4} + \frac{\delta}{2}\log(2u\sqrt{c}) - \phi_m(\delta)\right]W^2(u)\right\}$$

where we define the real functions R and ϕ by

$$\Gamma[\tfrac{1}{2}(x + 1 + iy)] = R_x(y)e^{i\phi_x(y)} \tag{3.40}$$

with $\phi_x(0) = 0$ and

$$W^2(u) = 1 + \sum_{j=1} \frac{W^2_j(u)}{(\sqrt{c})^j}$$
$$= \left(1 - \frac{u}{2\sqrt{c}}\right)^{m/2}\left[\sum_{r=0} \frac{\{\frac{1}{2}(m+1+i\delta\}_r\{\frac{1}{2}(1-m+i\delta)\}_r}{r!\,(-2iu\sqrt{c})^r}\right.$$
$$\left(1 + \sum_{j=1}\frac{1}{c^j}\sum_{k=1}^{j} K^j_k(-2iu\sqrt{c})^k\right) + \frac{(m+2)(m+1)}{\frac{1}{2}(m+1-i\delta)} \tag{3.41}$$
$$\left.\cdot \sum_{r=0} \frac{\{\frac{1}{2}(m+3+i\delta)\}_r\{-\frac{1}{2}(m+1-i\delta)\}_r}{r!\,(-2iu\sqrt{c})^r}\sum_{j=1}\frac{1}{c^j}\sum_{k=1}^{j} L^j_k(-2iu\sqrt{c})^k\right].$$

Here $\{a\}_\alpha = \Gamma(a+\alpha)/\Gamma(a)$.

If now we require

$$c + \delta\log(2\sqrt{c}) - \phi_m(\delta) = (m+1)\frac{\pi}{4} + l\frac{\pi}{2} + 2\pi p \tag{3.42}$$

with p an arbitrary integer, the exponential factors in (3.37) and (3.39) become identical, and apart from factors independent of u, these two series then have the same lead terms for large c. The coefficients of these lead terms can be determined as asymptotic series in c^{-1} by extracting the terms of $W^1_j(u)$ and $W^2_j(u)$ that are independent of u, much as was done in Sections 2.4 and 2.6. We content ourselves with the lead terms of these series in c^{-1}. The ratio of the lead term of ψ^6 to that of ψ^7 is

$$k_7 = \frac{c^{(m+1)/2}R_m(\delta)e^{\delta(\pi/4)}}{2\Gamma(m+1)}\left[1 + O\left(\frac{1}{c}\right)\right] \tag{3.43}$$

and we take $k_7\psi^7$ to be an extension of the solution ψ^6.

3 5 *Solution for $x > 1 + c^{-1}$* When

$$\psi_{m,n}(x) = \frac{\exp\left\{i\left[cx + \frac{\delta}{2}\log\left(\frac{x+1}{x-1}\right)\right]\right\}}{\sqrt{x^2-1}}\,w(x)$$

and (3.1) are substituted into (1.1), Eq. (3.30) results once more. We multiply this solution by a suitably chosen complex constant and take the real part for ψ^9 as indicated in (1.29) where the M's and N's are as in Section 3.3. As $x \to \infty$, (1.29) gives formally

$$\psi^9_{m,n}(x) \sim \frac{\cos[cx - (n+1)\pi/2]}{cx} \tag{3.44}$$

so that for the connection with Flammer's radial function [2] p. 31 we have

$$R^{(1)}_{m,n}(c, x) = \psi^9_{m,n}(x) \tag{3.45}$$

whence (1.34).

3.6 *Joining of the Solution* $\psi^8_{m,n}$ *and* $\psi^9_{m,n}$. The situation here is very similar to that treated in Section 3.4. Substituting $x = 1 + uc^{-\frac{1}{2}}$ into (1.29) and expanding in powers of $1/\sqrt{c}$ we find

$$\psi^9_{m,n}(x = 1 + uc^{-\frac{1}{2}}) \sim \frac{c^{-3/4}\operatorname{Re}\exp\left\{i\left[c + u\sqrt{c} - \frac{\delta}{2}\log\frac{u}{2\sqrt{c}} - (n+1)\frac{\pi}{2}\right]\right\}}{\sqrt{2u}} \cdot \left[1 + \sum_{j=1} \frac{W^1_j(-u)}{(\sqrt{c})^j}\right]. \tag{3.46}$$

Similarly ψ^8 becomes

$$\psi^8_{m,n}(x = 1 + uc^{-\frac{1}{2}}) \sim \frac{\sqrt{2}\Gamma(m+1)e^{\delta(\pi/4)}}{\sqrt{u}c^{(2m+1)/4}R_m(\delta)} \cdot \operatorname{Re}\exp\left\{i\left[u\sqrt{c} - (m+1)\frac{\pi}{4} - \frac{\delta}{2}\log(2u\sqrt{c}) + \phi_m(\delta)\right]\right\} W^2(-u) \tag{3.47}$$

with $W^2(u)$ given by (3.41) Comparing these results we find that if

$$c + \delta\log(2\sqrt{c}) - \phi_m(\delta) = (n+1)\pi/2 - (m+1)\pi/4 + 2\pi q \tag{3.48}$$

where q is an arbitrary integer, then the ratio,

$$r = \frac{2\Gamma(m+1)e^{\delta(\pi/4)}}{c^{(m-1)/2}R_m(\delta)}\left[1 + O\left(\frac{1}{c}\right)\right], \tag{3.49}$$

of the lead term of (3.47) to the lead term of (3.46) is independent of u We treat $r\psi^9$ as the continuation of ψ^8. From (3.49), (1.32) and (1.23), Eq. (1.33) then follows.

3.7 *Determination of* δ. By matching various solutions, we have arrived at two equations, (3.42) and (3.48) that δ must satisfy. Comparison of these equations shows that we must have $p = q$. We henceforth take (3.48) as the fundamental restriction on δ arising from matching solutions.

We now write this equation in the form $y_1(\delta) = y_2(\delta)$ where

$$\begin{aligned} y_1 &= \delta\log(2\sqrt{c}) + c - \pi/4 - h(m, n, q) \\ y_2 &= \phi_m(\delta), \qquad h = n\pi/2 - m\pi/4 + 2\pi q \end{aligned} \tag{3.50}$$

and fix the definition of $\phi_x(y)$ given in (3.40) by taking $\phi_x(0) = 0$. Figure 3 shows plots of y_2 vs δ for $m = 0, 1, 2, 3, 4$ and for $h = 0$ plots of members of the family of straight lines (3.50) corresponding to $c = .17, .25, .5, 1.0$. The envelope $y = (\delta/2)\log(-2\delta/e) - \pi/4$, $\delta \leqq 0$ of this family of straight lines is also shown. Solutions of (3.48) for given m and n can be visualized by translating the family of straight lines vertically by an amount h and observing the intersection of the appropriate straight line and $y_2 = \phi_m$ curve. For given m and n, there are infinitely many h values corresponding to the possible choices of q. Since for large

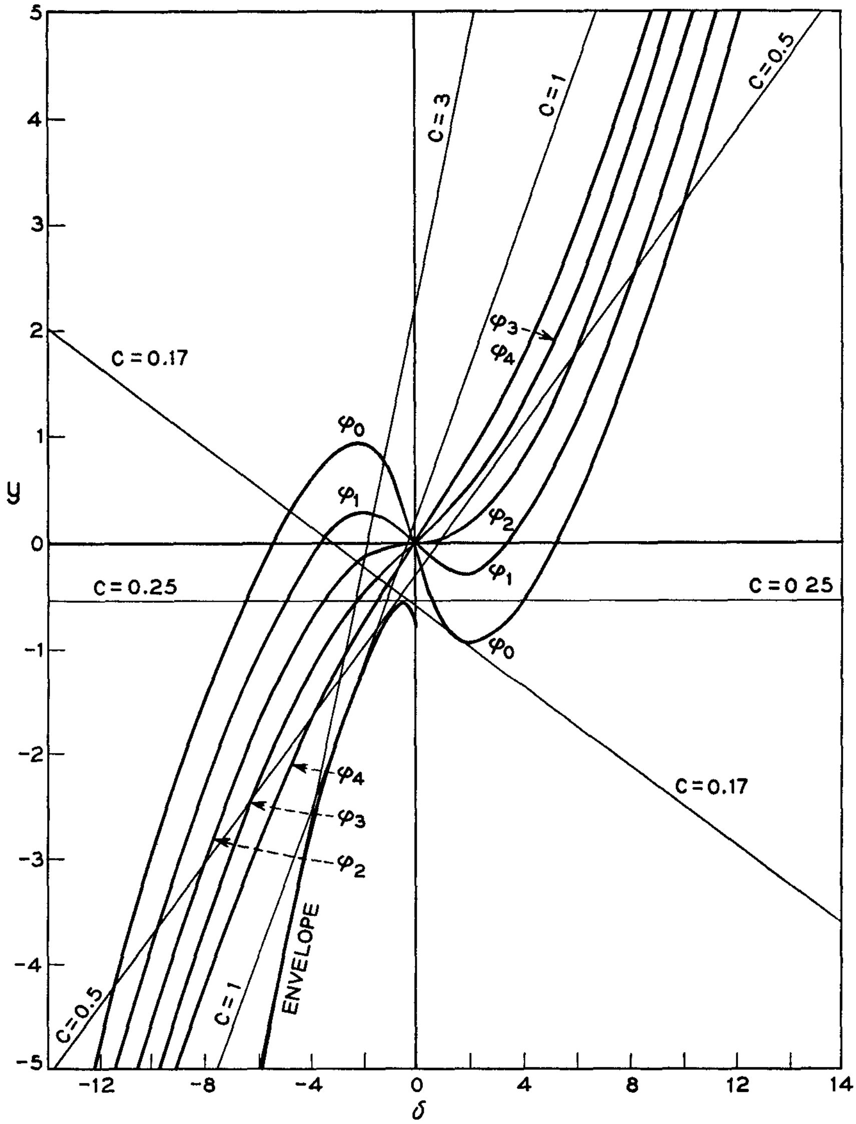

FIG. 3. Curves for determining δ

$|\delta|$, $\phi_m(\delta) \sim (\delta/2)\log(|\delta|/2) + m\pi/4$, there will always be a real intersection of the straight line and the $y_2 = \phi$ curve.

We thus see that for given values of c, m and n, there will in general be many values of δ and q for which (3.48) is satisfied. Once such a value of δ has been chosen, however, by (1.23)–(1.28) and (1.32) a formal solution $\psi_{m,n}(x)$ to (1.1) is completely determined valid for $|x| \leqq 1$. From general theory, this solution must have precisely $l = n - m$ zeros in the open interval $|x| < 1$. This, then, is the awkward criterion for choosing the proper values of δ and q to satisfy (3.48), n, c and m being given: the resultant solution must have precisely $n - m$ zeros in $|x| < 1$.

The situation for small or moderate values of c is quite complicated. Theoretical determination of the number of zeros contributed to ψ by ψ^6 and ψ^7 as a function of δ and c seems hopeless when more than one term in either of the series (1.24) or (1.27) must be taken into account. Numerical work for $m = 0$ and small values of n and c indicates that the correct root is always obtained by taking $q = 0$ and choosing the δ root of smallest absolute value.

The situation is more amenable to analysis when c is assumed so large that the zeros of ψ are completely determined by the lead terms of (1.24) and (1.27). For large enough c then, we have

$$\psi = \begin{cases} (1 - x^2)^{-\frac{1}{2}} \begin{smallmatrix}\sin\\ \cos\end{smallmatrix} \left(cx + \dfrac{\delta}{2}\log\dfrac{1+x}{1-x}\right), & 0 \leqq x \leqq 1 - \dfrac{1}{\sqrt{c}} \\ k_7(1 - x^2)^{m/2} e^{ic(1-x)} \Phi(\beta, m + 1, -2ic(1 - x)), & \\ & 1 - \dfrac{1}{\sqrt{c}} \leqq x \leqq 1, \end{cases} \tag{3.51}$$

and we can now estimate the number, Z, of zeros of (3.51) for $|x| \leqq 1$ as c becomes large. We assume $\delta = 0(1)$. From the first of (3.51), as x goes from $-1 + c^{-\frac{1}{2}}$ to $1 - c^{-\frac{1}{2}}$, the argument of the trigonometric function changes by an amount

$$2\left[c - \sqrt{c} + \frac{\delta}{2}\log 2\sqrt{c} + \frac{\delta}{2}\log\left(1 - \frac{1}{2\sqrt{c}}\right)\right]$$

so that the number of zeros, Z_1, contributed by this interval is

$$Z_1 = \frac{2}{\pi}\left[c - \sqrt{c} - \frac{\delta}{2}\log 2\sqrt{c}\right] + O(c^{-\frac{1}{2}}). \tag{3.52}$$

The number of zeros, Z_2, contributed by the second line of (3.51) is twice the number of zeros of the function $f(t) = e^{it}\Phi(\beta, m + 1, -2it)$ in the interval $0 \leqq t \leqq \sqrt{c}$. Now for large t,

$$f(t) \sim \frac{2\Gamma(m + 1)e^{-\delta(\pi/4)}}{(2t)^{(m+1)/2}R_m(\delta)} \cos\left[t + \frac{\delta}{2}\log 2t - \phi_m(\delta) - (m + 1)\frac{\pi}{4}\right]$$

([16] Vol. I p. 278). Assuming that for all a and b greater than some fixed number M, the number of zeros of $f(t)$ for $a < t < b$ differs from the number of zeros of

$\cos[t + \delta/2 \log 2t - \phi_m(\delta) - (m+1)\pi/4]$ in the same internal by no more than two, we find

$$Z_2 = \frac{2}{\pi}\left[\sqrt{c} + \frac{\delta}{2}\log 2\sqrt{c}\right] + O(1) \tag{3.53}$$

Combining these statements we have

$$Z = Z_1 + Z_2 = 2\pi^{-1}[c + \delta \log 2\sqrt{c}] + O(1) = n - m.$$

Comparison of this equation with (3 48) shows that for large c we must have

$$\phi_m(\delta) + (m+1)\pi/4 + 2\pi q = O(1). \tag{3.54}$$

In view of the foregoing, a useful limiting result can now be established. Let

$$n - m = [2\pi^{-1}(c + b \log 2\sqrt{c})]$$

where b is a fixed number (independent of c) and the brackets denote "integer part of". Equation (3.48) now becomes

$$(\delta - b)\log(2\sqrt{c}) = \phi_m(\delta) + (m+1)\pi/4 + 2\pi q + \epsilon$$

with $|\epsilon| < 1$ In virtue of (3 54), we then have the following result If

$$l = n - m = [2\pi^{-1}(c + b \log 2\sqrt{c})], \tag{3.55}$$

then

$$\lim_{c\to\infty} \delta = b. \tag{3.56}$$

IV. Asymptotics of λ_n, 4 1 *Fixed n and Large c.* We recall from [9], Eq. (27), that

$$\lambda_n = 2c\pi^{-1}[R_{0n}^{(1)}(c, 1)]^2 \tag{4.1}$$

where R is Flammer's radial prolate function. From the results of Section 2.8, it follows that, for fixed n and large c, $R_{0n}^{(1)}(c, x) \sim \sqrt{\pi/2c}\,\psi_{0,n}^4(x)$. From (2.64), one finds $\psi_{0,n}^4(1) = 1$, so that from (4.1) we obtain the weak result $\lambda_n \to 1$ as $c \to \infty$.

To obtain information about the rate at which $\lambda_n \to 1$, we return to (1 2) and (1 3) and differentiate the latter with respect to c. Multiply the result by $\psi_{0,n}(x)$ and integrate to find

$$\frac{\partial\lambda_n}{\partial c}\int_{-1}^{1}[\psi_{0,n}(x)]^2\,dx + \lambda_n\int_{-1}^{1}\psi_{0,n}(x)\,\frac{\partial\psi_{0,n}(x)}{\partial c}\,dx$$
$$= \frac{1}{\pi}\int_{-1}^{1}dx\int_{-1}^{1}dy\,\cos c(x-y)\psi_{0,n}(x)\psi_{0,n}(y) + \lambda_n\int_{-1}^{1}\psi_{0,n}(y)\,\frac{\partial\psi_{0,n}(y)}{\partial c}\,dy.$$

Here, to obtain the last term, we have interchanged orders of integration and again used (1.3). We now have

$$\frac{\partial\lambda_n}{\partial c} = \frac{1}{\pi}\left[\int_{-1}^{1}dx\,\cos cx\psi_{0,n}(x)\right]^2 + \frac{1}{\pi}\left[\int_{-1}^{1}dx\,\sin cx\psi_{0,n}(x)\right]^2 \tag{4.2}$$

where we have assumed ψ real and normalized so that $\int_{-1}^{1} \psi^2\, dx = 1$. Suppose now n is even. The last integral in (4.2) then vanishes. However, for even n, (1.2) gives

$$\alpha_n \psi_{0,n}(1) = \int_{-1}^{1} \cos cy \psi_{0,n}(y)\, dy,$$

so that (4.2) becomes $\partial\lambda_n/\partial c = |\alpha_n|^2[\psi_{0,n}(1)]^2/\pi$. The same result is obtained when n is odd. Using the relationship between λ_n and α_n given in (1.3), we have

$$\frac{1}{\lambda_n}\frac{\partial\lambda_n}{\partial c} = \frac{2}{c}[\psi_{0,n}(1)]^2 \tag{4.3}$$

a relationship also found by Fuchs [17].

Now from (1.4), (1.8) and (1.16), $[\psi_{0,n}(1)]^2 = N_{m,n}^2 k_3^2$. Using the expansion given in (1.12), (1.14), (1.15) and (1.16), the right side of (4.3) can be expressed in the form

$$\frac{2^{3n+3}\sqrt{\pi}}{n!} c^{n+\frac{1}{2}} e^{-2c}\left[1 + \sum_1 \frac{\kappa_j}{c^j}\right].$$

Integrating (4.3) then yields

$$-\log\lambda_n = \frac{2^{3n+3}\sqrt{\pi}}{n!}\int_c^\infty dt\, t^{n+\frac{1}{2}} e^{-2t}\left[1 + \sum_1 \frac{\kappa_j}{t^j}\right].$$

Working out the first few κ's and making obvious expansions, we find finally

$$\begin{aligned}\lambda_n = 1 - \frac{2^{3n+2}\sqrt{\pi}c^{n+\frac{1}{2}}e^{-2c}}{n!} \times \Big[1 &- \frac{6n^2 - 2n + 3}{32c} \\ &- \frac{114n^5 - 192n^4 + 1056n^3 - 3454n^2 + 873n + 2334}{2^{11}3!\, c^2} + O(c^{-3})\Big].\end{aligned} \tag{4.4}$$

The lead term in the expansion of $1 - \lambda_n$ has been given previously by Fuchs [17].

4.2 *Fixed n and Small c.* For small c, it is not hard to derive that

$$\begin{aligned}\psi_{0,n}(x) = P_n(x) + c^2\Big[&\frac{n(n-1)}{2(2n-1)^2(2n+1)} P_{n-2}(x) \\ &- \frac{(n+1)(n+2)}{2(2n+3)^2(2n+1)} P_{n+2}(x)\Big] + O(c^4)\end{aligned} \tag{4.5}$$

is a solution of (1.1). From [9] Eq. (40),

$$\lambda_{n+1}/\lambda_n = -\left[\int_{-1}^{1} \psi_{0,n}'(x)\psi_{0,n+1}(x)\, dx\right] \Big/ \int_{-1}^{1} \psi_{0,n}(x)\psi_{0,n+1}'(x)\, dx.$$

Using (4.5) in this equation, one finds

$$\lambda_{n+1} = \lambda_n \frac{(n+1)c^2}{(2n+1)^2(2n+3)^2}[1 + O(c^2)]. \tag{4.6}$$

From (1.2)

$$\alpha_0 \psi_{0,0}(1) = \int_{-1}^{1} \cos cy\psi_{0,0}(y)\, dy. \tag{4.7}$$

From (4.5) and (4.7), one finds immediately that $\alpha_0 = 2 + O(c^2)$, whence from (1.3) $\lambda_0 = 2\pi^{-1}c[1 + O(c^2)]$. Combined with (4.6), this gives finally

$$\lambda_n = \frac{2}{\pi}\left[\frac{2^{2n}(n!)^3}{(2n)!\,(2n+1)!}\right]^2 c^{2n+1}[1 + O(c^2)]. \tag{4.8}$$

We now use (4.3) to improve this estimate. From (4.5) one has

$$\int_{-1}^{1} [\psi_{0,n}(x)]^2\, dx = \frac{2}{2n+1} + O(c^4).$$

This result, combined with (4.5), gives

$$\overset{*}{\psi}_{0,n}(1) = \sqrt{\frac{2n+1}{2}}\left[1 - \frac{c^2}{(2n-1)^2(2n+3)^2} + O(c^4)\right]$$

where $\overset{*}{\psi}_{0,n}$ is a solution of (1.1) normalized to have square integral unity in $(-1, 1)$. Using this value in (4.3) and integrating from $c = 0$, we find (1.35). Here we have used (4.8) to determine the constant of integration.

4.3. *Asymptotics for n and c Both Large.* Equations (3.45) and (3.49) can be combined to yield

$$R_{0,n}^{(1)}(c, 1) = \frac{R_0(\delta)e^{-\delta\pi/4}}{2\sqrt{c}}\, \overset{8}{\psi}_{0,n}(1)[1 + O(c^{-1})]. \tag{4.9}$$

From (1.26) we see that $\overset{8}{\psi}_{0,n}(1) = 1$. Equations (4.1) and (4.9) thus give

$$\lambda_n = \frac{1}{2\pi} R_0^2(\delta)e^{-\delta\pi/2}[1 + O(c^{-1})]. \tag{4.10}$$

Recall now from (3.40) that $R_0(\delta)$ is defined by

$$\Gamma\left(\frac{1}{2} + i\frac{\delta}{2}\right) = R_0(\delta)e^{i\varphi_0(\delta)}.$$

Multiply this equation by its complex conjugate. There results

$$[R_0(\delta)]^2 = \Gamma\left(\frac{1}{2} + i\frac{\delta}{2}\right)\Gamma\left(\frac{1}{2} - i\frac{\delta}{2}\right) = \frac{\pi}{\cosh \delta(\pi/2)}. \tag{4.11}$$

Here we have used the functional relation ([16], Vol. 1, p. 3, Eq. (7)) for the gamma function

$$\Gamma(\tfrac{1}{2} + z)\Gamma(\tfrac{1}{2} - z) = \pi \sec \pi z$$

Equations (4.10) and (4.11) then give

$$\lambda_n = [1 + e^{\pi\delta}]^{-1}[1 + O(c^{-1})]. \tag{4.12}$$

This formula together with (3.55) and (3.56) yields the limit statement for fixed b reported as (1.39) and (1.40).

As pointed out in Section 1.5, (4.12) is remarkably accurate even for small n and c when δ is chosen as the root of smallest absolute value of (3.48) with $q = 0$.

REFERENCES

[1] J MEIXNER AND F W SCHAFKE, "Mathieusche Funktionen und Spharoidfunktionen", Springer-Verlag, Berlin, 1954

[2] C FLAMMER, "Spheroidal Wave Functions", Stanford Univ Press, Stanford, 1957

[3] J A STRATTON, P M MORSE, L J CHU, J D C LITTLE, AND P J CORBATÓ, "Spheroidal Wave Functions", John Wiley & Sons, Inc , New York, 1956

[4] D. SLEPIAN, "Estimation of Signal Parameters in the Presence of Noise", IRE Trans **PGIT-3**, pp 68–89, 1954

[5] G D BOYD AND J P. GORDON, "Confocal Multimode Resonator for Millimeter Through Optical Wavelength Masers", Bell System J , **40**, pp 489–508, 1961

[6] F J DYSON, "Statistical Theory of Energy Levels of Complex Systems, III", J of Math Phys , **3**, pp 166–175, 1962

[7] M GAUDIN, "Sur la loi limite de l'espacement des valeurs propres d'une matrice aléatorre", Nuclear Phys , **25**, pp 447–458, 1961

[8] D R RHODES, "The Optimum Line Source for the Best Mean-Square Approximation to a Given Radiation Pattern", IEEE Trans on Antennas and Propagation, **AP-11**, pp 440–446, 1963.

[9] D SLEPIAN AND H O POLLAK, "Prolate Spheroidal Wave Functions, Fourier Analysis and Uncertainty—I", Bell System Tech J , **40**, pp 43–64, 1961

[10] H J LANDAU AND H O POLLAK, "Prolate Spheroidal Wave Functions, Fourier Analysis and Uncertainty—II", Bell System Tech J , **40**, pp 65–84, 1961

[11] H J LANDAU AND H O POLLAK, "Prolate Spheroidal Wave Functions, Fourier Analysis and Uncertainty—III", Bell System Tech J , **41**, pp 1295-1336, 1962

[12] H J LANDAU, "The Eigenvalue Behavior of Certain Convolution Equations", Trans Am Math Soc., to appear

[13] M PETRICH, "On the Number of Orthogonal Signals which can be Placed in a WT-Product", J Soc Ind Appl Math , **11**, pp 936–940, 1963

[14] R SIPS, "Représéntation Asymptotique des Fonctions de Mathieu et des Fonctions d'Onde Sphéroidales", Trans Am Math Soc , **66**, pp 93–134, 1949

[15] H J W MULLER, "Asymptotic Expansions of Prolate Spheroidal Wave Functions and Their Characteristic Numbers", J Reine u Ange Math , **212**, pp 26–48, 1963

[16] A ERDÉLYI, "Higher Transcendental Functions", McGraw-Hill Book Co , New York, 1953

[17] W H. J FUCHS, "On the Eigenvalues of an Integral Equation Arising in the Theory of Band-limited Signals," J Math Analysis and Appl , **9**, pp 317–330, 1964

[18] W S BROWN, "The ALPAK System for Nonnumerical Algebra on a Digital Computer —I", Bell System Tech J , **42**, pp 2081–2119, 1963
W S BROWN, J P HYDE AND B A TAGUE, "The ALPAK System for Nonnumerical Algebra on a Digital Computer—II", Bell System Tech J , **43**, pp 785–804, 1964

[19] H WIDOM, "Asymptotic Behavior of the Eigenvalues of Certain Integral Equations II," Archive for Rational Analysis and Mechanics, **17**, pp 215–229, 1964

[20] D SLEPIAN, "Prolate Spheroidal Wave Functions, Fourier Analysis and Uncertainty—IV Extensions to Many Dimensions, Generalized Prolate Spheroidal Functions," Bell System Tech J , **43**, pp 3009-3057, 1964

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